
NASA-GLENN CHEMICAL EQUILIBRIUM PROGRAM CEA2, FEBRUARY 5, 2004
 BY BONNIE MCBRIDE AND SANFORD GORDON
 REFS: NASA RP-1311, PART I, 1994 AND NASA RP-1311, PART II, 1996

CEA analysis performed on Thu 01-Jan-2026 10:12:05

Problem Type: "Rocket" (Infinite Area Combustor)

prob case = kaw00_____8420 ro equilibrium

Pressure (1 value):

p,psia= 120

Chamber/Exit Pressure Ratio (1 value):

pi/p= 620.297

Oxidizer/Fuel Wt. ratio (21 values):

o/f = 0.5, 0.6, 0.7, 0.8, 0.9, 1, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2
 , 2.1, 2.2, 2.3, 2.4, 2.5

You selected the following fuels and oxidizers:

reac

fuel C2H8N2(L),UDMH wt%= 50.0000

fuel N2H4 wt%= 50.0000

oxid N2O4 wt%=100.0000

You selected these options for output:

long version of output

Proportions of any products will be expressed as Mass Fractions.

output massf

Heat will be expressed as siunits

output siunits

Plot parameters:

output plot isp ae

Input prepared by this script:/var/www/sites/cearun/cgi-bin/CEARUN/prepareInputFile.cgi

IMPORTANT: The following line is the end of your CEA input file!
 end

OPTIONS: TP=F HP=F SP=F TV=F UV=F SV=F DETN=F SHOCK=F REFL=F INCD=F
 RKT=T FROZ=F EQL=T IONS=F SIUNIT=T DEBUGF=F SHKDBG=F DETDBG=F TRNSPT=F

TRACE= 0.00E+00 S/R= 0.000000E+00 H/R= 0.000000E+00 U/R= 0.000000E+00

Pc,BAR = 8.273677

Pc/P = 620.2970

SUBSONIC AREA RATIOS =

SUPERSONIC AREA RATIOS =

NFZ= 1 Mdot/Ac= 0.000000E+00 Ac/At= 0.000000E+00

REACTANT	WT.FRAC	(ENERGY/R),K	TEMP,K	DENSITY
EXPLODED FORMULA				
F: C2H8N2(L),UDMH	0.500000	0.588128E+04	298.15	0.0000
C 2.00000 H 8.00000 N 2.00000				
F: N2H4	0.500000	0.000000E+00	0.00	0.0000

N 2.00000 H 4.00000
 O: N204 1.000000 0.000000E+00 0.00 0.0000
 N 2.00000 O 4.00000

SPECIES BEING CONSIDERED IN THIS SYSTEM
 (CONDENSED PHASE MAY HAVE NAME LISTED SEVERAL TIMES)
 LAST thermo.inp UPDATE: 9/09/04

g 7/97 *C	tpis79 *CH	g 4/02 CH2
g 4/02 CH3	g11/00 CH2OH	g 7/00 CH30
g 8/99 CH4	g 7/00 CH3OH	srd 01 CH300H
g 8/99 *CN	g12/99 CNN	tpis79 *CO
g 9/99 *C02	tpis91 C0OH	tpis91 *C2
g 6/01 C2H	g 1/91 C2H2,acetylene	g 5/01 C2H2,vinylidene
g 4/02 CH2C0,ketene	g 3/02 O(CH)20	srd 01 HO(C0)20H
g 7/01 C2H3,vinyl	g 9/00 CH3CN	g 6/96 CH3C0,acetyl
g 1/00 C2H4	g 8/88 C2H40,ethylen-o	g 8/88 CH3CH0,ethanal
g 6/00 CH3C0OH	srd 01 OHCH2C0OH	g 7/00 C2H5
g 7/00 C2H6	g 8/88 CH3N2CH3	g 8/88 C2H50H
g 7/00 CH3OCH3	srd 01 CH3O2CH3	g 7/00 CCN
tpis91 CNC	srd 01 OCCN	tpis79 C2N2
g 8/00 C2O	tpis79 *C3	n 4/98 C3H3,1-propyn1
n 4/98 C3H3,2-propyn1	g 2/00 C3H4,allene	g 1/00 C3H4,propyne
g 5/90 C3H4,cyclo-	g 3/01 C3H5,allyl	g 2/00 C3H6,propylene
g 1/00 C3H6,cyclo-	g 6/01 C3H60,propylox	g 6/97 C3H60,acetone
g 1/02 C3H60,propanal	g 7/01 C3H7,n-propyl	g 9/85 C3H7,i-propyl
g 2/00 C3H8	g 2/00 C3H80,1propanol	g 2/00 C3H80,2propanol
srd 01 CNCOCN	g 7/88 C302	g tpis *C4
g 7/01 C4H2,butadiyne	g 8/00 C4H4,1,3-cyclo-	n10/92 C4H6,butadiene
n10/93 C4H6,1butyne	n10/93 C4H6,2butyne	g 8/00 C4H6,cyclo-
n 4/88 C4H8,1-butene	n 4/88 C4H8,cis2-buten	n 4/88 C4H8,tr2-butene
n 4/88 C4H8,isobutene	g 8/00 C4H8,cyclo-	g10/00 (CH3C0OH)2
n10/84 C4H9,n-butyl	n10/84 C4H9,i-butyl	g 1/93 C4H9,s-butyl
g 1/93 C4H9,t-butyl	g12/00 C4H10,n-butane	g 8/00 C4H10,isobutane
g 6/01 C4N2	g 8/00 *C5	g 5/90 C5H6,1,3cyclo-
g 1/93 C5H8,cyclo-	n 4/87 C5H10,1-pentene	g 2/01 C5H10,cyclo-
n10/84 C5H11,pentyl	g 1/93 C5H11,t-pentyl	n10/85 C5H12,n-pentane
n10/85 C5H12,i-pentane	n10/85 CH3C(CH3)2CH3	g 2/93 C6H2
g11/00 C6H5,phenyl	g 8/00 C6H50,phenoxy	g 8/00 C6H6
g 8/00 C6H5OH,phenol	g 1/93 C6H10,cyclo-	n 4/87 C6H12,1-hexene
g 6/90 C6H12,cyclo-	n10/83 C6H13,n-hexyl	g 6/01 C6H14,n-hexane
g 7/01 C7H7,benzyl	g 1/93 C7H8	g12/00 C7H80,cresol-mx
n 4/87 C7H14,1-heptene	n10/83 C7H15,n-heptyl	n10/85 C7H16,n-heptane
n10/85 C7H16,2-methylh	n 4/89 C8H8,styrene	n10/86 C8H10,ethylbenz
n 4/87 C8H16,1-octene	n10/83 C8H17,n-octyl	n 4/85 C8H18,n-octane
n 4/85 C8H18,isooctane	n10/83 C9H19,n-nonyl	g 3/01 C10H8,naphthale
n10/83 C10H21,n-decyl	G12/12 C11H21	g 8/00 C12H9,o-bipheny
g 8/00 C12H10,biphenyl	g 6/97 *H	g 6/01 HCN
g 1/01 HCO	tpis89 HCCN	g 6/01 HCCO
g 6/01 HNC	g 7/00 HNC0	g10/01 HNO
tpis89 HN02	g 5/99 HN03	g 4/02 HO2
tpis78 *H2	g 5/01 HCHO,formaldehy	g 6/01 HCOOH
g 8/89 H2O	g 6/99 H2O2	g 6/01 (HCOOH)2
g 5/97 *N	g 6/01 NCO	g 4/99 *NH
g 3/01 NH2	tpis89 NH3	tpis89 NH2OH
tpis89 *NO	g 4/99 NO2	j12/64 NO3
tpis78 *N2	J12/64 N2O	g 6/01 NCN
g 5/99 N2H2	tpis89 NH2NO2	g 4/99 N2H4
g 4/99 N2O3	tpis89 N2O4	g 4/99 N2O5
tpis89 N3	g 4/99 N3H	g 5/97 *O
g 4/02 *OH	tpis89 *O2	g 8/01 O3
g 12/0 THDCPD,endo	g 12/0 THDCPD,exo	g11/99 N2H4(L)
n 4/83 C(gr)	n 4/83 C(gr)	n 4/83 C(gr)
n12/84 CH3OH(L)	n12/84 C2H5OH(L)	n 4/85 C6H14(L),n-hexa
n12/88 C6H5NH2(L)	n10/86 C6H6(L)	g11/99 H2O(cr)
g 8/01 H2O(L)	g 8/01 H2O(L)	

O/F = 0.500000

ENTHALPY (KG-MOL)(K)/KG	EFFECTIVE FUEL h(2)/R 0.48930525E+02	EFFECTIVE OXIDANT h(1)/R 0.00000000E+00	MIXTURE h0/R 0.32620350E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.11092933E-01
*H	0.12896952E+00	0.00000000E+00	0.85979681E-01
*N	0.47845361E-01	0.21736532E-01	0.39142418E-01
*O	0.00000000E+00	0.43473063E-01	0.14491021E-01

POINT ITN T C H N O

Pinf/Pt = 1.834260

Pinf/Pt = 1.801132

Pinf/Pt = 1.802211

Pinf/Pt = 1.802167

ADD C(gr)

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA

CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 0.50000 %FUEL= 66.666667 R,EQ.RATIO= 4.497662 PHI,EQ.RATIO= 4.497662

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.8022	620.30
P, BAR	8.2737	4.5910	0.01334
T, K	1330.29	1179.46	632.48
RHO, KG/CU M	1.0192 0	6.4355-1	4.2010-3
H, KJ/KG	271.22	-176.67	-2947.53
U, KJ/KG	-540.59	-890.05	-3265.03
G, KJ/KG	-18997.6	-17260.8	-12108.8
S, KJ/(KG)(K)	14.4847	14.4847	14.4847

M, (1/n)	13.625	13.747	16.563
MW, MOL WT	13.625	13.747	15.721
(dLV/dLP)t	-1.00639	-1.01993	-1.05537
(dLV/dLT)p	1.0652	1.2332	1.9381
Cp, KJ/(KG)(K)	2.8978	4.1149	10.3193
GAMMAS	1.3030	1.2557	1.1459
SON VEL,M/SEC	1028.5	946.5	603.2
MACH NUMBER	0.000	1.000	4.206

PERFORMANCE PARAMETERS

Ae/At	1.0000	57.143
CSTAR, M/SEC	1358.4	1358.4
CF	0.6968	1.8678
Ivac, M/SEC	1700.2	2662.4
Isp, M/SEC	946.5	2537.2

MASS FRACTIONS

CH4	0.00195	0.00717	0.08053
*CO	0.29510	0.28034	0.00958

*CO2	0.01914	0.02803	0.10982
HCN	0.00003	0.00001	0.00000
*H2	0.07991	0.07835	0.04795
H2O	0.05559	0.05781	0.16499
NH3	0.00023	0.00024	0.00005
*N2	0.54805	0.54805	0.54822
C(gr)	0.00000	0.00000	0.03886

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH3OH	CH300H	*CN	CNN
COOH	*C2	C2H	C2H2, acetylene	C2H2, vinylidene
CH2CO, ketene	O(CH)2O	HO(CO)2OH	C2H3, vinyl	CH3CN
CH3CO, acetyl	C2H4	C2H4O, ethylene-o	CH3CHO, ethanal	CH3COOH
OHCH2COOH	C2H5	C2H6	CH3N2CH3	C2H5OH
CH3OCH3	CH3O2CH3	CCN	CNC	OCCN
C2N2	C2O	*C3	C3H3, 1-propynyl	C3H3, 2-propynyl
C3H4, allene	C3H4, propyne	C3H4, cyclo-	C3H5, allyl	C3H6, propylene
C3H6, cyclo-	C3H6O, propylox	C3H6O, acetone	C3H6O, propanal	C3H7, n-propyl
C3H7, i-propyl	C3H8	C3H8O, 1propanol	C3H8O, 2propanol	CNCOCN
C3O2	*C4	C4H2, butadiyne	C4H4, 1,3-cyclo-	C4H6, butadiene
C4H6, 1butyne	C4H6, 2butyne	C4H6, cyclo-	C4H8, 1-butene	C4H8, cis2-buten
C4H8, tr2-butene	C4H8, isobutene	C4H8, cyclo-	(CH3COOH)2	C4H9, n-butyl
C4H9, i-butyl	C4H9, s-butyl	C4H9, t-butyl	C4H10, n-butane	C4H10, isobutane
C4N2	*C5	C5H6, 1,3cyclo-	C5H8, cyclo-	C5H10, 1-pentene
C5H10, cyclo-	C5H11, pentyl	C5H11, t-pentyl	C5H12, n-pentane	C5H12, i-pentane
CH3C(CH3)2CH3	C6H2	C6H5, phenyl	C6H5O, phenoxy	C6H6
C6H5OH, phenol	C6H10, cyclo-	C6H12, 1-hexene	C6H12, cyclo-	C6H13, n-hexyl
C6H14, n-hexane	C7H7, benzyl	C7H8	C7H8O, cresol-mx	C7H14, 1-heptene
C7H15, n-heptyl	C7H16, n-heptane	C7H16, 2-methylh	C8H8, styrene	C8H10, ethylbenz
C8H16, 1-octene	C8H17, n-octyl	C8H18, n-octane	C8H18, isooctane	C9H19, n-nonyl
C10H8, naphthale	C10H21, n-decyl	C11H21	C12H9, o-bipheny	C12H10, biphenyl
*H	HCO	HCCN	HCCO	HNC
HNCO	HNO	HNO2	HNO3	H02
HCHO, formaldehy	HCOOH	H2O2	(HCOOH)2	*N
NCO	*NH	NH2	NH2OH	*NO
NO2	NO3	N2O	NCN	N2H2
NH2NO2	N2H4	N2O3	N2O4	N2O5
N3	N3H	*O	*OH	*O2
O3	THDCPD, endo	THDCPD, exo	N2H4(L)	CH3OH(L)
C2H5OH(L)	C6H14(L), n-hexa	C6H5NH2(L)	C6H6(L)	H2O(cr)
H2O(L)				

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

O/F = 0.600000

ENTHALPY (KG-MOL)(K)/KG	EFFECTIVE FUEL h(2)/R	EFFECTIVE OXIDANT h(1)/R	MIXTURE h0/R
	0.48930525E+02	0.00000000E+00	0.30581578E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.10399625E-01
*H	0.12896952E+00	0.00000000E+00	0.80605951E-01
*N	0.47845361E-01	0.21736532E-01	0.38054550E-01
*O	0.00000000E+00	0.43473063E-01	0.16302399E-01

POINT ITN	T	C	H	N	O
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Pinf/Pt = 1.836715

Pinf/Pt = 1.840629

ADD C(gr)

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA

CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 0.60000 %FUEL= 62.500000 R,EQ.RATIO= 3.748051 PHI,EQ.RATIO= 3.748051

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.8406	620.30
P, BAR	8.2737	4.4950	0.01334
T, K	1573.99	1362.42	650.24
RHO, KG/CU M	9.0678-1	5.6927-1	4.1161-3
H, KJ/KG	254.27	-264.13	-3119.64
U, KJ/KG	-658.16	-1053.74	-3443.69
G, KJ/KG	-22407.3	-19879.5	-12481.5
S, KJ/(KG)(K)	14.3975	14.3975	14.3975
M, (1/n)	14.343	14.346	16.684
MW, MOL WT	14.343	14.346	15.939
(dLV/dLP)t	-1.00021	-1.00062	-1.05145
(dLV/dLT)p	1.0015	1.0058	1.8867
Cp, KJ/(KG)(K)	2.4725	2.4527	10.0335
GAMMAS	1.3071	1.3131	1.1433
SON VEL,M/SEC	1092.1	1018.2	608.7
MACH NUMBER	0.000	1.000	4.268

PERFORMANCE PARAMETERS

Ae/At	1.0000	54.212
CSTAR, M/SEC	1427.3	1427.3
CF	0.7134	1.8199
Ivac, M/SEC	1793.7	2722.4
Isp, M/SEC	1018.2	2597.7

MASS FRACTIONS

CH4	0.00003	0.00015	0.05605
*CO	0.27651	0.27110	0.01766
*CO2	0.02311	0.03131	0.15285
HCN	0.00001	0.00001	0.00000
*H2	0.07037	0.07071	0.04956
H2O	0.09692	0.09369	0.15720
NH3	0.00009	0.00010	0.00004
*N2	0.53294	0.53293	0.53299
C(gr)	0.00000	0.00000	0.03365

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH3OH	CH3OOH	*CN	CNN

C00H	*C2	C2H	C2H2, acetylene	C2H2, vinylidene
CH2CO, ketene	O(CH)2O	H0(CO)2OH	C2H3, vinyl	CH3CN
CH3CO, acetyl	C2H4	C2H4O, ethylene-o	CH3CHO, ethanal	CH3C00H
OHCH2C00H	C2H5	C2H6	CH3N2CH3	C2H5OH
CH3OCH3	CH3O2CH3	CCN	CNC	OCCN
C2N2	C2O	*C3	C3H3, 1-propyn1	C3H3, 2-propyn1
C3H4, allene	C3H4, propyne	C3H4, cyclo-	C3H5, allyl	C3H6, propylene
C3H6, cyclo-	C3H6O, propylox	C3H6O, acetone	C3H6O, propanal	C3H7, n-propyl
C3H7, i-propyl	C3H8	C3H8O, 1propanol	C3H8O, 2propanol	CNCOCN
C3O2	*C4	C4H2, butadiyne	C4H4, 1,3-cyclo-	C4H6, butadiene
C4H6, 1butyne	C4H6, 2butyne	C4H6, cyclo-	C4H8, 1-butene	C4H8, cis2-buten
C4H8, tr2-butene	C4H8, isobutene	C4H8, cyclo-	(CH3C00H)2	C4H9, n-butyl
C4H9, i-butyl	C4H9, s-butyl	C4H9, t-butyl	C4H10, n-butane	C4H10, isobutane
C4N2	*C5	C5H6, 1,3cyclo-	C5H8, cyclo-	C5H10, 1-pentene
C5H10, cyclo-	C5H11, pentyl	C5H11, t-pentyl	C5H12, n-pentane	C5H12, i-pentane
CH3C(CH3)2CH3	C6H2	C6H5, phenyl	C6H5O, phenoxy	C6H6
C6H5OH, phenol	C6H10, cyclo-	C6H12, 1-hexene	C6H12, cyclo-	C6H13, n-hexyl
C6H14, n-hexane	C7H7, benzyl	C7H8	C7H8O, cresol-mx	C7H14, 1-heptene
C7H15, n-heptyl	C7H16, n-heptane	C7H16, 2-methylh	C8H8, styrene	C8H10, ethylbenz
C8H16, 1-octene	C8H17, n-octyl	C8H18, n-octane	C8H18, isooctane	C9H19, n-nonyl
C10H8, naphthale	C10H21, n-decyl	C11H21	C12H9, o-bipheny	C12H10, biphenyl
*H	HCO	HCCN	HCCO	HNC
HNCO	HNO	HNO2	HNO3	H02
HCHO, formaldehy	HCOOH	H2O2	(HCOOH)2	*N
NCO	*NH	NH2	NH2OH	*NO
NO2	NO3	N2O	NCN	N2H2
NH2NO2	N2H4	N2O3	N2O4	N2O5
N3	N3H	*O	*OH	*O2
O3	THDCPD, endo	THDCPD, exo	N2H4(L)	CH3OH(L)
C2H5OH(L)	C6H14(L), n-hexa	C6H5NH2(L)	C6H6(L)	H2O(cr)
H2O(L)				

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

$$O/F = 0.700000$$

ENTHALPY	EFFECTIVE FUEL	EFFECTIVE OXIDANT	MIXTURE
(KG-MOL)(K)/KG	h(2)/R	h(1)/R	h0/R
	0.48930525E+02	0.00000000E+00	0.28782662E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.97878825E-02
*H	0.12896952E+00	0.00000000E+00	0.75864425E-01
*N	0.47845361E-01	0.21736532E-01	0.37094666E-01
*O	0.00000000E+00	0.43473063E-01	0.17900673E-01

POINT ITN T C H N O

$$P_{inf}/P_t = 1.826614$$

$$P_{inf}/P_t = 1.832257$$

ADD C(gr)

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

$$P_{in} = 120.0 \text{ PSIA}$$

$$\text{CASE} = \text{kaw00}$$

REACTANT	WT FRACTION	ENERGY	TEMP
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		(SEE NOTE)	KJ/KG-MOL	K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 0.70000 %FUEL= 58.823529 R,EQ.RATIO= 3.212615 PHI,EQ.RATIO= 3.212615

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.8323	620.30
P, BAR	8.2737	4.5156	0.01334
T, K	1807.73	1574.97	662.53
RHO, KG/CU M	8.3067-1	5.2038-1	4.0938-3
H, KJ/KG	239.31	-324.24	-3310.58
U, KJ/KG	-756.71	-1191.98	-3636.40
G, KJ/KG	-25525.9	-22771.9	-12753.4
S, KJ/(KG)(K)	14.2528	14.2528	14.2528

M, (1/n)	15.090	15.091	16.907
MW, MOL WT	15.090	15.091	16.385
(dLV/dLP)t	-1.00007	-1.00006	-1.04873
(dLV/dLT)p	1.0009	1.0004	1.8600
Cp, KJ/(KG)(K)	2.4516	2.3956	9.9326
GAMMAS	1.2904	1.2989	1.1397
SON VEL,M/SEC	1133.7	1061.7	609.4
MACH NUMBER	0.000	1.000	4.373

PERFORMANCE PARAMETERS

Ae/At	1.0000	50.647
CSTAR, M/SEC	1497.6	1497.6
CF	0.7089	1.7792
Ivac, M/SEC	1879.0	2786.8
Isp, M/SEC	1061.7	2664.5

MASS FRACTIONS

CH4	0.00000	0.00000	0.04200
*CO	0.25759	0.25313	0.02641
*CO2	0.02601	0.03303	0.19119
*H	0.00001	0.00000	0.00000
HCN	0.00001	0.00000	0.00000
*H2	0.06129	0.06161	0.04924
H2O	0.13551	0.13264	0.14897
NH3	0.00004	0.00004	0.00003
*N2	0.51953	0.51954	0.51955
C(gr)	0.00000	0.00000	0.02261

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH3OH	CH3OOH	*CN	CNN
COOH	*C2	C2H	C2H2,acetylene	C2H2,vinylidene
CH2CO,ketene	O(CH)2O	HO(CO)2OH	C2H3,vinyl	CH3CN
CH3CO,acetyl	C2H4	C2H4O,ethylen-o	CH3CHO,ethanal	CH3COOH
OHCH2COOH	C2H5	C2H6	CH3N2CH3	C2H5OH
CH3OCH3	CH3O2CH3	CCN	CNC	OCCN
C2N2	C2O	*C3	C3H3,1-propyn1	C3H3,2-propyn1
C3H4,allene	C3H4,propyne	C3H4,cyclo-	C3H5,allyl	C3H6,propylene
C3H6,cyclo-	C3H6O,propylox	C3H6O,acetone	C3H6O,propanal	C3H7,n-propyl
C3H7,i-propyl	C3H8	C3H8O,1propanol	C3H8O,2propanol	CNCOCN
C3O2	*C4	C4H2,butadiyne	C4H4,1,3-cyclo-	C4H6,butadiene
C4H6,1butyne	C4H6,2butyne	C4H6,cyclo-	C4H8,1-butene	C4H8,cis2-buten
C4H8,tr2-butene	C4H8,isobutene	C4H8,cyclo-	(CH3COOH)2	C4H9,n-butyl
C4H9,i-butyl	C4H9,s-butyl	C4H9,t-butyl	C4H10,n-butane	C4H10,isobutane
C4N2	*C5	C5H6,1,3cyclo-	C5H8,cyclo-	C5H10,1-pentene

C5H10,cyclo-	C5H11,pentyl	C5H11,t-pentyl	C5H12,n-pentane	C5H12,i-pentane
CH3C(CH3)2CH3	C6H2	C6H5,phenyl	C6H5O,phenoxy	C6H6
C6H5OH,phenol	C6H10,cyclo-	C6H12,1-hexene	C6H12,cyclo-	C6H13,n-hexyl
C6H14,n-hexane	C7H7,benzyl	C7H8	C7H8O,cresol-mx	C7H14,1-heptene
C7H15,n-heptyl	C7H16,n-heptane	C7H16,2-methylh	C8H8,styrene	C8H10,ethylbenz
C8H16,1-octene	C8H17,n-octyl	C8H18,n-octane	C8H18,isooctane	C9H19,n-nonyl
C10H8,naphthale	C10H21,n-decyl	C11H21	C12H9,o-bipheny	C12H10,biphenyl
HCO	HCCN	HCCO	HNC	HNCO
HNO	HNO2	HNO3	HO2	HCHO,formaldehy
HCOOH	H2O2	(HCOOH)2	*N	NCO
*NH	NH2	NH2OH	*NO	NO2
NO3	N2O	NCN	N2H2	NH2NO2
N2H4	N2O3	N2O4	N2O5	N3
N3H	*O	*OH	*O2	O3
THDCPD,endo	THDCPD,exo	N2H4(L)	CH3OH(L)	C2H5OH(L)
C6H14(L),n-hexa	C6H5NH2(L)	C6H6(L)	H2O(cr)	H2O(L)

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

O/F = 0.800000

ENTHALPY	EFFECTIVE FUEL	EFFECTIVE OXIDANT	MIXTURE
(KG-MOL)(K)/KG	h(2)/R	h(1)/R	h0/R
	0.48930525E+02	0.00000000E+00	0.27183625E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.92441112E-02
*H	0.12896952E+00	0.00000000E+00	0.71649735E-01
*N	0.47845361E-01	0.21736532E-01	0.36241437E-01
*O	0.00000000E+00	0.43473063E-01	0.19321361E-01

POINT ITN	T	C	H	N	O
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Pinf/Pt = 1.816778

Pinf/Pt = 1.823426

ADD C(gr)

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA

CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 0.80000 %FUEL= 55.555556 R,EQ.RATIO= 2.811038 PHI,EQ.RATIO= 2.811038

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.8234	620.30
P, BAR	8.2737	4.5374	0.01334
T, K	2022.82	1774.38	670.88
RHO, KG/CU M	7.7835-1	4.8671-1	4.1089-3
H, KJ/KG	226.02	-372.59	-3506.85
U, KJ/KG	-836.96	-1304.86	-3831.47
G, KJ/KG	-28255.2	-25355.8	-12952.8

S, KJ/(KG)(K)	14.0800	14.0800	14.0800
M, (1/n)	15.822	15.825	17.183
MW, MOL WT	15.822	15.825	16.943
(dLV/dLP)t	-1.00014	-1.00005	-1.04700
(dLV/dLT)p	1.0031	1.0008	1.8497
Cp, KJ/(KG)(K)	2.4553	2.3770	9.9240
GAMMAS	1.2742	1.2843	1.1361
SON VEL,M/SEC	1163.8	1094.2	607.3
MACH NUMBER	0.000	1.000	4.499

PERFORMANCE PARAMETERS

Ae/At	1.0000	47.435
CSTAR, M/SEC	1553.6	1553.6
CF	0.7043	1.7587
Ivac, M/SEC	1946.2	2851.2
Isp, M/SEC	1094.2	2732.3

MASS FRACTIONS

CH4	0.00000	0.00000	0.03342
*CO	0.24052	0.23665	0.03453
*CO2	0.02892	0.03499	0.22457
*H	0.00003	0.00001	0.00000
*H2	0.05320	0.05350	0.04792
H2O	0.16969	0.16722	0.14201
NH3	0.00003	0.00002	0.00003
*N2	0.50760	0.50760	0.50760
*OH	0.00002	0.00000	0.00000
C(gr)	0.00000	0.00000	0.00992

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH3OH	CH3OOH	*CN	CNN
COOH	*C2	C2H	C2H2,acetylene	C2H2,vinylidene
CH2CO,ketene	O(CH)2O	HO(CO)2OH	C2H3,vinyl	CH3CN
CH3CO,acetyl	C2H4	C2H4O,ethylen-o	CH3CHO,ethanal	CH3COOH
OHCH2COOH	C2H5	C2H6	CH3N2CH3	C2H5OH
CH3OCH3	CH3O2CH3	CCN	CNC	OCCN
C2N2	C2O	*C3	C3H3,1-propyn1	C3H3,2-propyn1
C3H4,allene	C3H4,propyne	C3H4,cyclo-	C3H5,allyl	C3H6,propylene
C3H6,cyclo-	C3H6O,propylox	C3H6O,acetone	C3H6O,propanal	C3H7,n-propyl
C3H7,i-propyl	C3H8	C3H8O,1propanol	C3H8O,2propanol	CNCOCN
C3O2	*C4	C4H2,butadiyne	C4H4,1,3-cyclo-	C4H6,butadiene
C4H6,1butyne	C4H6,2butyne	C4H6,cyclo-	C4H8,1-butene	C4H8,cis2-buten
C4H8,tr2-butene	C4H8,isobutene	C4H8,cyclo-	(CH3COOH)2	C4H9,n-butyl
C4H9,i-butyl	C4H9,s-butyl	C4H9,t-butyl	C4H10,n-butane	C4H10,isobutane
C4N2	*C5	C5H6,1,3cyclo-	C5H8,cyclo-	C5H10,1-pentene
C5H10,cyclo-	C5H11,pentyl	C5H11,t-pentyl	C5H12,n-pentane	C5H12,i-pentane
CH3C(CH3)2CH3	C6H2	C6H5,phenyl	C6H5O,phenoxy	C6H6
C6H5OH,phenol	C6H10,cyclo-	C6H12,1-hexene	C6H12,cyclo-	C6H13,n-hexyl
C6H14,n-hexane	C7H7,benzyl	C7H8	C7H8O,cresol-mx	C7H14,1-heptene
C7H15,n-heptyl	C7H16,n-heptane	C7H16,2-methylh	C8H8,styrene	C8H10,ethylbenz
C8H16,1-octene	C8H17,n-octyl	C8H18,n-octane	C8H18,isoctane	C9H19,n-nonyl
C10H8,naphthale	C10H21,n-decyl	C11H21	C12H9,o-bipheny	C12H10,biphenyl
HCN	HCO	HCCN	HCCO	HNC
HNCO	HNO	HNO2	HNO3	HO2
HCHO,formaldehy	HCOOH	H2O2	(HCOOH)2	*N
NCO	*NH	NH2	NH2OH	*NO
NO2	NO3	N2O	NCN	N2H2
NH2NO2	N2H4	N2O3	N2O4	N2O5
N3	N3H	*O	*O2	O3
THDCPD,endo	THDCPD,exo	N2H4(L)	CH3OH(L)	C2H5OH(L)

C6H14(L),n-hexa C6H5NH2(L) C6H6(L) H2O(cr) H2O(L)

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

O/F = 0.900000

ENTHALPY (KG-MOL)(K)/KG	EFFECTIVE FUEL h(2)/R 0.48930525E+02	EFFECTIVE OXIDANT h(1)/R 0.00000000E+00	MIXTURE h0/R 0.25752908E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.87575791E-02
*H	0.12896952E+00	0.00000000E+00	0.67878696E-01
*N	0.47845361E-01	0.21736532E-01	0.35478021E-01
*O	0.00000000E+00	0.43473063E-01	0.20592504E-01

POINT ITN	T	C	H	N	O
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Pinf/Pt = 1.806477

Pinf/Pt = 1.814810

Pinf/Pt = 1.814861

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA

CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 0.90000 %FUEL= 52.631579 R,EQ.RATIO= 2.498701 PHI,EQ.RATIO= 2.498701

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.8149	620.30
P, BAR	8.2737	4.5588	0.01334
T, K	2219.26	1960.63	677.02
RHO, KG/CU M	7.4140-1	4.6265-1	4.1422-3
H, KJ/KG	214.12	-411.52	-3701.30
U, KJ/KG	-901.84	-1396.90	-4023.31
G, KJ/KG	-30622.0	-27654.0	-13108.3
S, KJ/(KG)(K)	13.8947	13.8947	13.8947
M, (1/n)	16.535	16.544	17.481
MW, MOL WT	16.535	16.544	17.481
(dLV/dLP)t	-1.00038	-1.00011	-1.04417
(dLV/dLT)p	1.0092	1.0027	1.7994
Cp, KJ/(KG)(K)	2.4978	2.3765	9.1700
GAMMAS	1.2573	1.2699	1.1412
SON VEL,M/SEC	1184.5	1118.6	606.2
MACH NUMBER	0.000	1.000	4.616

PERFORMANCE PARAMETERS

Ae/At	1.00000	44.647
CSTAR, M/SEC	1598.7	1598.7

CF	0.6997	1.7504
Ivac, M/SEC	1999.5	2913.4
Isp, M/SEC	1118.6	2798.4

MASS FRACTIONS

CH4	0.00000	0.00000	0.02591
*CO	0.22481	0.22141	0.04034
*CO2	0.03219	0.03753	0.25096
*H	0.00008	0.00002	0.00000
*H2	0.04596	0.04626	0.04628
H2O	0.19992	0.19783	0.13958
NH3	0.00002	0.00001	0.00002
*NO	0.00001	0.00000	0.00000
*N2	0.49691	0.49692	0.49691
*OH	0.00011	0.00002	0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH3OH	CH3OOH	*CN	CNN
COOH	*C2	C2H	C2H2,acetylene	C2H2,vinylidene
CH2CO,ketene	O(CH)2O	HO(CO)2OH	C2H3,vinyl	CH3CN
CH3CO,acetyl	C2H4	C2H4O,ethylen-o	CH3CHO,ethanal	CH3COOH
OHCH2COOH	C2H5	C2H6	CH3N2CH3	C2H5OH
CH3OCH3	CH3O2CH3	CCN	CNC	OCCN
C2N2	C2O	*C3	C3H3,1-propyn1	C3H3,2-propyn1
C3H4,allene	C3H4,propyne	C3H4,cyclo-	C3H5,allyl	C3H6,propylene
C3H6,cyclo-	C3H6O,propylox	C3H6O,acetone	C3H6O,propanal	C3H7,n-propyl
C3H7,i-propyl	C3H8	C3H8O,1propanol	C3H8O,2propanol	CNCOCN
C3O2	*C4	C4H2,butadiyne	C4H4,1,3-cyclo-	C4H6,butadiene
C4H6,1butyne	C4H6,2butyne	C4H6,cyclo-	C4H8,1-butene	C4H8,cis2-buten
C4H8,tr2-butene	C4H8,isobutene	C4H8,cyclo-	(CH3COOH)2	C4H9,n-butyl
C4H9,i-butyl	C4H9,s-butyl	C4H9,t-butyl	C4H10,n-butane	C4H10,isobutane
C4N2	*C5	C5H6,1,3cyclo-	C5H8,cyclo-	C5H10,1-pentene
C5H10,cyclo-	C5H11,pentyl	C5H11,t-pentyl	C5H12,n-pentane	C5H12,i-pentane
CH3C(CH3)2CH3	C6H2	C6H5,phenyl	C6H5O,phenoxy	C6H6
C6H5OH,phenol	C6H10,cyclo-	C6H12,1-hexene	C6H12,cyclo-	C6H13,n-hexyl
C6H14,n-hexane	C7H7,benzyl	C7H8	C7H8O,cresol-mx	C7H14,1-heptene
C7H15,n-heptyl	C7H16,n-heptane	C7H16,2-methylh	C8H8,styrene	C8H10,ethylbenz
C8H16,1-octene	C8H17,n-octyl	C8H18,n-octane	C8H18,isoctane	C9H19,n-nonyl
C10H8,naphthale	C10H21,n-decyl	C11H21	C12H9,o-bipheny	C12H10,biphenyl
HCN	HCO	HCCN	HCCO	HNC
HNCO	HNO	HNO2	HNO3	HO2
HCHO,formaldehy	HCOOH	H2O2	(HCOOH)2	*N
NCO	*NH	NH2	NH2OH	NO2
NO3	N2O	NCN	N2H2	NH2NO2
N2H4	N2O3	N2O4	N2O5	N3
N3H	*O	*O2	O3	THDCPD,endo
THDCPD,exo	N2H4(L)	C(gr)	CH3OH(L)	C2H5OH(L)
C6H14(L),n-hexa	C6H5NH2(L)	C6H6(L)	H2O(cr)	H2O(L)

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

O/F = 1.000000

ENTHALPY	EFFECTIVE FUEL	EFFECTIVE OXIDANT	MIXTURE
(KG-MOL)(K)/KG	h(2)/R	h(1)/R	h0/R
	0.48930525E+02	0.00000000E+00	0.24465262E+02

KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.83197001E-02
*H	0.12896952E+00	0.00000000E+00	0.64484761E-01
*N	0.47845361E-01	0.21736532E-01	0.34790946E-01
*O	0.00000000E+00	0.43473063E-01	0.21736532E-01

POINT	ITN	T	C	H	N	O
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Pinf/Pt = 1.795522

Pinf/Pt = 1.805794

Pinf/Pt = 1.805875

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA

CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 1.00000 %FUEL= 50.000000 R,EQ.RATIO= 2.248831 PHI,EQ.RATIO= 2.248831

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.8059	620.30
P, BAR	8.2737	4.5815	0.01334
T, K	2396.20	2133.82	685.96
RHO, KG/CU M	7.1524-1	4.4531-1	4.1600-3
H, KJ/KG	203.42	-442.06	-3888.92
U, KJ/KG	-953.35	-1470.90	-4209.55
G, KJ/KG	-32638.5	-29687.9	-13290.6
S, KJ/(KG)(K)	13.7059	13.7059	13.7059

M, (1/n)	17.223	17.244	17.788
MW, MOL WT	17.223	17.244	17.788
(dLV/dLP)t	-1.00091	-1.00029	-1.03381
(dLV/dLT)p	1.0213	1.0075	1.6047
Cp, KJ/(KG)(K)	2.5953	2.4070	7.3087
GAMMAS	1.2393	1.2548	1.1506
SON VEL,M/SEC	1197.3	1136.2	607.4
MACH NUMBER	0.000	1.000	4.710

PERFORMANCE PARAMETERS

Ae/At	1.0000	42.513
CSTAR, M/SEC	1635.2	1635.2
CF	0.6948	1.7495
Ivac, M/SEC	2041.7	2973.0
Isp, M/SEC	1136.2	2860.9

MASS FRACTIONS

CH4	0.00000	0.00000	0.01395
*CO	0.21012	0.20710	0.04150
*CO2	0.03600	0.04075	0.26268
*H	0.00018	0.00006	0.00000
*H2	0.03944	0.03976	0.04472
H2O	0.22652	0.22493	0.14984
NH3	0.00001	0.00001	0.00002
*NO	0.00003	0.00001	0.00000

*N2 0.48728 0.48730 0.48729
 *OH 0.00041 0.00009 0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
 WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH3OH	CH3OOH	*CN	CNN
COOH	*C2	C2H	C2H2,acetylene	C2H2,vinylidene
CH2CO,ketene	O(CH)2O	HO(CO)2OH	C2H3,vinyl	CH3CN
CH3CO,acetyl	C2H4	C2H4O,ethylen-o	CH3CHO,ethanal	CH3COOH
OHCH2COOH	C2H5	C2H6	CH3N2CH3	C2H5OH
CH3OCH3	CH3O2CH3	CCN	CNC	OCCN
C2N2	C2O	*C3	C3H3,1-propynl	C3H3,2-propynl
C3H4,allene	C3H4,propyne	C3H4,cyclo-	C3H5,allyl	C3H6,propylene
C3H6,cyclo-	C3H6O,propylox	C3H6O,acetone	C3H6O,propanal	C3H7,n-propyl
C3H7,i-propyl	C3H8	C3H8O,1propanol	C3H8O,2propanol	CNCOCN
C3O2	*C4	C4H2,butadiyne	C4H4,1,3-cyclo-	C4H6,butadiene
C4H6,1butyne	C4H6,2butyne	C4H6,cyclo-	C4H8,1-butene	C4H8,cis2-buten
C4H8,tr2-butene	C4H8,isobutene	C4H8,cyclo-	(CH3COOH)2	C4H9,n-butyl
C4H9,i-butyl	C4H9,s-butyl	C4H9,t-butyl	C4H10,n-butane	C4H10,isobutane
C4N2	*C5	C5H6,1,3cyclo-	C5H8,cyclo-	C5H10,1-pentene
C5H10,cyclo-	C5H11,pentyl	C5H11,t-pentyl	C5H12,n-pentane	C5H12,i-pentane
CH3C(CH3)2CH3	C6H2	C6H5,phenyl	C6H5O,phenoxy	C6H6
C6H5OH,phenol	C6H10,cyclo-	C6H12,1-hexene	C6H12,cyclo-	C6H13,n-hexyl
C6H14,n-hexane	C7H7,benzyl	C7H8	C7H8O,cresol-mx	C7H14,1-heptene
C7H15,n-heptyl	C7H16,n-heptane	C7H16,2-methylh	C8H8,styrene	C8H10,ethylbenz
C8H16,1-octene	C8H17,n-octyl	C8H18,n-octane	C8H18,isoctane	C9H19,n-nonyl
C10H8,naphthale	C10H21,n-decyl	C11H21	C12H9,o-bipheny	C12H10,biphenyl
HCN	HCO	HCCN	HCCO	HNC
HNCO	HNO	HN02	HN03	H02
HCHO,formaldehy	HCOOH	H2O2	(HCOOH)2	*N
NCO	*NH	NH2	NH2OH	N02
N03	N2O	NCN	N2H2	NH2N02
N2H4	N2O3	N2O4	N2O5	N3
N3H	*O	*O2	O3	THDCPD,endo
THDCPD,exo	N2H4(L)	C(gr)	CH3OH(L)	C2H5OH(L)
C6H14(L),n-hexa	C6H5NH2(L)	C6H6(L)	H2O(cr)	H2O(L)

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

O/F = 1.100000

ENTHALPY (KG-MOL)(K)/KG	EFFECTIVE FUEL h(2)/R 0.48930525E+02	EFFECTIVE OXIDANT h(1)/R 0.00000000E+00	MIXTURE h0/R 0.23300250E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.79235239E-02
*H	0.12896952E+00	0.00000000E+00	0.61414058E-01
*N	0.47845361E-01	0.21736532E-01	0.34169307E-01
*O	0.00000000E+00	0.43473063E-01	0.22771604E-01

POINT ITN	T	C	H	N	O
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Pinf/Pt = 1.784322

Pinf/Pt = 1.796039

Pinf/Pt = 1.796154

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA

CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 1.10000 %FUEL= 47.619048 R,EQ.RATIO= 2.044392 PHI,EQ.RATIO= 2.044392

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.7962	620.30
P, BAR	8.2737	4.6063	0.01334
T, K	2552.65	2293.22	701.44
RHO, KG/CU M	6.9712-1	4.3301-1	4.1511-3
H, KJ/KG	193.73	-465.02	-4066.46
U, KJ/KG	-993.10	-1528.81	-4387.78
G, KJ/KG	-34313.9	-31465.6	-13548.7
S, KJ/(KG)(K)	13.5183	13.5183	13.5183

M, (1/n)	17.883	17.924	18.150
MW, MOL WT	17.883	17.924	18.150
(dLV/dLP)t	-1.00184	-1.00070	-1.01736
(dLV/dLT)p	1.0414	1.0171	1.3045
Cp, KJ/(KG)(K)	2.7587	2.4830	4.6563
GAMMA _s	1.2209	1.2385	1.1765
SON VEL,M/SEC	1203.8	1147.8	614.9
MACH NUMBER	0.000	1.000	4.747

PERFORMANCE PARAMETERS

Ae/At	1.0000	41.019
CSTAR, M/SEC	1664.6	1664.6
CF	0.6895	1.7535
Ivac, M/SEC	2074.6	3029.0
Isp, M/SEC	1147.8	2919.0

MASS FRACTIONS

CH4	0.00000	0.00000	0.00497
*CO	0.19618	0.19346	0.04264
*CO2	0.04046	0.04474	0.26809
*H	0.00034	0.00013	0.00000
*H2	0.03356	0.03391	0.04237
H2O	0.24961	0.24881	0.16333
NH3	0.00001	0.00001	0.00002
*NO	0.00013	0.00003	0.00000
*N2	0.47853	0.47858	0.47859
*O	0.00001	0.00000	0.00000
*OH	0.00115	0.00033	0.00000
*O2	0.00001	0.00000	0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH3OH	CH3OOH	*CN	CNN
COOH	*C2	C2H	C2H2,acetylene	C2H2,vinylidene
CH2CO,ketene	O(CH)2O	HO(CO)2OH	C2H3,vinyl	CH3CN
CH3CO,acetyl	C2H4	C2H4O,ethylen-o	CH3CHO,ethanal	CH3COOH

OHCH ₂ COOH	C ₂ H ₅	C ₂ H ₆	CH ₃ N ₂ CH ₃	C ₂ H ₅ OH
CH ₃ OCH ₃	CH ₃ O ₂ CH ₃	CCN	CNC	OCCN
C ₂ N ₂	C ₂ O	*C ₃	C ₃ H ₃ ,1-propyn1	C ₃ H ₃ ,2-propyn1
C ₃ H ₄ ,allene	C ₃ H ₄ ,propyne	C ₃ H ₄ ,cyclo-	C ₃ H ₅ ,allyl	C ₃ H ₆ ,propylene
C ₃ H ₆ ,cyclo-	C ₃ H ₆ O,propylox	C ₃ H ₆ O,acetone	C ₃ H ₆ O,propanal	C ₃ H ₇ ,n-propyl
C ₃ H ₇ ,i-propyl	C ₃ H ₈	C ₃ H ₈ O,1propanol	C ₃ H ₈ O,2propanol	CNCOCN
C ₃ O ₂	*C ₄	C ₄ H ₂ ,butadiyne	C ₄ H ₄ ,1,3-cyclo-	C ₄ H ₆ ,butadiene
C ₄ H ₆ ,1butyne	C ₄ H ₆ ,2butyne	C ₄ H ₆ ,cyclo-	C ₄ H ₈ ,1-butene	C ₄ H ₈ ,cis2-buten
C ₄ H ₈ ,tr2-butene	C ₄ H ₈ ,isobutene	C ₄ H ₈ ,cyclo-	(CH ₃ COOH) ₂	C ₄ H ₉ ,n-butyl
C ₄ H ₉ ,i-butyl	C ₄ H ₉ ,s-butyl	C ₄ H ₉ ,t-butyl	C ₄ H ₁₀ ,n-butane	C ₄ H ₁₀ ,isobutane
C ₄ N ₂	*C ₅	C ₅ H ₆ ,1,3cyclo-	C ₅ H ₈ ,cyclo-	C ₅ H ₁₀ ,1-pentene
C ₅ H ₁₀ ,cyclo-	C ₅ H ₁₁ ,pentyl	C ₅ H ₁₁ ,t-pentyl	C ₅ H ₁₂ ,n-pentane	C ₅ H ₁₂ ,i-pentane
CH ₃ C(CH ₃) ₂ CH ₃	C ₆ H ₂	C ₆ H ₅ ,phenyl	C ₆ H ₅ O,phenoxy	C ₆ H ₆
C ₆ H ₅ OH,phenol	C ₆ H ₁₀ ,cyclo-	C ₆ H ₁₂ ,1-hexene	C ₆ H ₁₂ ,cyclo-	C ₆ H ₁₃ ,n-hexyl
C ₆ H ₁₄ ,n-hexane	C ₇ H ₇ ,benzyl	C ₇ H ₈	C ₇ H ₈ O,cresol-mx	C ₇ H ₁₄ ,1-heptene
C ₇ H ₁₅ ,n-heptyl	C ₇ H ₁₆ ,n-heptane	C ₇ H ₁₆ ,2-methylh	C ₈ H ₈ ,styrene	C ₈ H ₁₀ ,ethylbenz
C ₈ H ₁₆ ,1-octene	C ₈ H ₁₇ ,n-octyl	C ₈ H ₁₈ ,n-octane	C ₈ H ₁₈ ,isooctane	C ₉ H ₁₉ ,n-nonyl
C ₁₀ H ₈ ,naphthale	C ₁₀ H ₂₁ ,n-decyl	C ₁₁ H ₂₁	C ₁₂ H ₉ ,o-bipheny	C ₁₂ H ₁₀ ,biphenyl
HCN	HCO	HCCN	HCCO	HNC
HNCO	HNO	HN ₂	HN ₃	HO ₂
HCHO,formaldehy	HC ₂ OOH	H ₂ O ₂	(HCOOH) ₂	*N
NCO	*NH	NH ₂	NH ₂ OH	N ₂
NO ₃	N ₂ O	NCN	N ₂ H ₂	NH ₂ N ₂
N ₂ H ₄	N ₂ O ₃	N ₂ O ₄	N ₂ O ₅	N ₃
N ₃ H	O ₃	THDCPD,endo	THDCPD,exo	N ₂ H ₄ (L)
C(gr)	CH ₃ OH(L)	C ₂ H ₅ OH(L)	C ₆ H ₁₄ (L),n-hexa	C ₆ H ₅ NH ₂ (L)
C ₆ H ₆ (L)	H ₂ O(cr)	H ₂ O(L)		

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

$$O/F = 1.200000$$

ENTHALPY (KG-MOL)(K)/KG	EFFECTIVE FUEL h(2)/R	EFFECTIVE OXIDANT h(1)/R	MIXTURE h0/R
	0.48930525E+02	0.00000000E+00	0.22241148E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.75633637E-02
*H	0.12896952E+00	0.00000000E+00	0.58622510E-01
*N	0.47845361E-01	0.21736532E-01	0.33604181E-01
*O	0.00000000E+00	0.43473063E-01	0.23712580E-01

POINT ITN T C H N O

$$P_{inf}/P_t = 1.773460$$

$$P_{inf}/P_t = 1.785629$$

$$P_{inf}/P_t = 1.785768$$

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

$$P_{in} = 120.0 \text{ PSIA}$$

$$\text{CASE} = \text{kaw00}$$

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C ₂ H ₈ N ₂ (L),UDMH	0.5000000	48900.000	298.150
FUEL	N ₂ H ₄	0.5000000	0.000	0.000
OXIDANT	N ₂ O ₄	1.0000000	0.000	0.000

O/F= 1.20000 %FUEL= 45.454545 R,EQ.RATIO= 1.874026 PHI,EQ.RATIO= 1.874026

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.7858	620.30
P, BAR	8.2737	4.6331	0.01334
T, K	2688.20	2437.64	737.09
RHO, KG/CU M	6.8519-1	4.2467-1	4.0602-3
H, KJ/KG	184.92	-481.29	-4230.74
U, KJ/KG	-1022.57	-1572.27	-4559.25
G, KJ/KG	-35662.3	-32987.3	-14059.8
S, KJ/(KG)(K)	13.3351	13.3351	13.3351
M, (1/n)	18.510	18.578	18.655
MW, MOL WT	18.510	18.578	18.655
(dLV/dLP)t	-1.00326	-1.00142	-1.00261
(dLV/dLT)p	1.0711	1.0336	1.0438
Cp, KJ/(KG)(K)	2.9947	2.6179	2.4803
GAMMAS	1.2031	1.2213	1.2394
SON VEL,M/SEC	1205.3	1154.3	638.1
MACH NUMBER	0.000	1.000	4.657

PERFORMANCE PARAMETERS

Ae/At	1.0000	40.628
CSTAR, M/SEC	1687.8	1687.8
CF	0.6839	1.7607
Ivac, M/SEC	2099.4	3082.3
Isp, M/SEC	1154.3	2971.8

MASS FRACTIONS

CH4	0.00000	0.00000	0.00058
*CO	0.18278	0.18027	0.04510
*CO2	0.04568	0.04961	0.26041
*H	0.00053	0.00025	0.00000
*H2	0.02829	0.02862	0.03824
H2O	0.26914	0.26956	0.18498
NH3	0.00000	0.00000	0.00001
*NO	0.00036	0.00010	0.00000
*N2	0.47051	0.47063	0.47068
*O	0.00005	0.00001	0.00000
*OH	0.00261	0.00093	0.00000
*O2	0.00005	0.00001	0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH3OH	CH3OOH	*CN	CNN
COOH	*C2	C2H	C2H2,acetylene	C2H2,vinylidene
CH2CO,ketene	O(CH)2O	HO(CO)2OH	C2H3,vinyl	CH3CN
CH3CO,acetyl	C2H4	C2H4O,ethylen-o	CH3CHO,ethanal	CH3COOH
OHCH2COOH	C2H5	C2H6	CH3N2CH3	C2H5OH
CH3OCH3	CH3O2CH3	CCN	CNC	OCCN
C2N2	C2O	*C3	C3H3,1-propynl	C3H3,2-propynl
C3H4,allene	C3H4,propyne	C3H4,cyclo-	C3H5,allyl	C3H6,propylene
C3H6,cyclo-	C3H6O,propylox	C3H6O,acetone	C3H6O,propanal	C3H7,n-propyl
C3H7,i-propyl	C3H8	C3H8O,1propanol	C3H8O,2propanol	CNCOCN
C3O2	*C4	C4H2,butadiyne	C4H4,1,3-cyclo-	C4H6,butadiene
C4H6,1butyne	C4H6,2butyne	C4H6,cyclo-	C4H8,1-butene	C4H8,cis2-buten
C4H8,tr2-butene	C4H8,isobutene	C4H8,cyclo-	(CH3COOH)2	C4H9,n-butyl
C4H9,i-butyl	C4H9,s-butyl	C4H9,t-butyl	C4H10,n-butane	C4H10,isobutane
C4N2	*C5	C5H6,1,3cyclo-	C5H8,cyclo-	C5H10,1-pentene
C5H10,cyclo-	C5H11,pentyl	C5H11,t-pentyl	C5H12,n-pentane	C5H12,i-pentane
CH3C(CH3)2CH3	C6H2	C6H5,phenyl	C6H5O,phenoxy	C6H6

C6H5OH,phenol	C6H10,cyclo-	C6H12,1-hexene	C6H12,cyclo-	C6H13,n-hexyl
C6H14,n-hexane	C7H7,benzyl	C7H8	C7H8O,cresol-mx	C7H14,1-heptene
C7H15,n-heptyl	C7H16,n-heptane	C7H16,2-methylh	C8H8,styrene	C8H10,ethylbenz
C8H16,1-octene	C8H17,n-octyl	C8H18,n-octane	C8H18,isoctane	C9H19,n-nonyl
C10H8,naphthale	C10H21,n-decyl	C11H21	C12H9,o-bipheny	C12H10,biphenyl
HCN	HCO	HCCN	HCCO	HNC
HNCO	HNO	HNO2	HNO3	H02
HCHO,formaldehy	HCOOH	H2O2	(HCOOH)2	*N
NCO	*NH	NH2	NH2OH	N02
NO3	N2O	NCN	N2H2	NH2NO2
N2H4	N2O3	N2O4	N2O5	N3
N3H	O3	THDCPD,endo	THDCPD,exo	N2H4(L)
C(gr)	CH3OH(L)	C2H5OH(L)	C6H14(L),n-hexa	C6H5NH2(L)
C6H6(L)	H2O(cr)	H2O(L)		

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

O/F = 1.300000

ENTHALPY (KG-MOL)(K)/KG	EFFECTIVE FUEL h(2)/R 0.48930525E+02	EFFECTIVE OXIDANT h(1)/R 0.00000000E+00	MIXTURE h0/R 0.21274141E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.72345218E-02
*H	0.12896952E+00	0.00000000E+00	0.56073705E-01
*N	0.47845361E-01	0.21736532E-01	0.33088196E-01
*O	0.00000000E+00	0.43473063E-01	0.24571731E-01

POINT ITN	T	C	H	N	O
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Pinf/Pt = 1.763372
 Pinf/Pt = 1.774943
 Pinf/Pt = 1.775081

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA
 CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 1.30000 %FUEL= 43.478261 R,EQ.RATIO= 1.729870 PHI,EQ.RATIO= 1.729870

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.7751	620.30
P, BAR	8.2737	4.6610	0.01334
T, K	2802.99	2565.76	794.30
RHO, KG/CU M	6.7816-1	4.1953-1	3.8980-3
H, KJ/KG	176.88	-491.84	-4377.64
U, KJ/KG	-1043.14	-1602.85	-4719.82
G, KJ/KG	-36703.8	-34251.2	-14828.8
S, KJ/(KG)(K)	13.1576	13.1576	13.1576

M, (1/n)	19.102	19.202	19.301
MW, MOL WT	19.102	19.202	19.301
(dLV/dLP)t	-1.00524	-1.00257	-1.00013
(dLV/dLT)p	1.1118	1.0589	1.0021
Cp, KJ/(KG)(K)	3.3110	2.8247	2.0937
GAMMAS	1.1866	1.2038	1.2602
SON VEL,M/SEC	1203.2	1156.5	656.7
MACH NUMBER	0.000	1.000	4.596

PERFORMANCE PARAMETERS

Ae/At	1.0000	41.240
CSTAR, M/SEC	1705.3	1705.3
CF	0.6782	1.7699
Ivac, M/SEC	2117.2	3131.5
Isp, M/SEC	1156.5	3018.1

MASS FRACTIONS

CH4	0.00000	0.00000	0.00003
*CO	0.16975	0.16733	0.04823
*CO2	0.05167	0.05548	0.24254
*H	0.00072	0.00039	0.00000
*H2	0.02361	0.02388	0.03267
H2O	0.28497	0.28708	0.21308
*NO	0.00083	0.00028	0.00000
*N2	0.46306	0.46332	0.46345
*O	0.00017	0.00004	0.00000
*OH	0.00503	0.00215	0.00000
*O2	0.00018	0.00004	0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH3OH	CH3OOH	*CN	CNN
COOH	*C2	C2H	C2H2,acetylene	C2H2,vinylidene
CH2CO,ketene	O(CH)2O	HO(CO)2OH	C2H3,vinyl	CH3CN
CH3CO,acetyl	C2H4	C2H4O,ethylen-o	CH3CHO,ethanal	CH3COOH
OHCH2COOH	C2H5	C2H6	CH3N2CH3	C2H5OH
CH3OCH3	CH3O2CH3	CCN	CNC	OCCN
C2N2	C2O	*C3	C3H3,1-propynl	C3H3,2-propynl
C3H4,allene	C3H4,propyne	C3H4,cyclo-	C3H5,allyl	C3H6,propylene
C3H6,cyclo-	C3H6O,propylox	C3H6O,acetone	C3H6O,propanal	C3H7,n-propyl
C3H7,i-propyl	C3H8	C3H8O,1propanol	C3H8O,2propanol	CNCOCN
C3O2	*C4	C4H2,butadiyne	C4H4,1,3-cyclo-	C4H6,butadiene
C4H6,1butyne	C4H6,2butyne	C4H6,cyclo-	C4H8,1-butene	C4H8,cis2-buten
C4H8,tr2-butene	C4H8,isobutene	C4H8,cyclo-	(CH3COOH)2	C4H9,n-butyl
C4H9,i-butyl	C4H9,s-butyl	C4H9,t-butyl	C4H10,n-butane	C4H10,isobutane
C4N2	*C5	C5H6,1,3cyclo-	C5H8,cyclo-	C5H10,1-pentene
C5H10,cyclo-	C5H11,pentyl	C5H11,t-pentyl	C5H12,n-pentane	C5H12,i-pentane
CH3C(CH3)2CH3	C6H2	C6H5,phenyl	C6H5O,phenoxy	C6H6
C6H5OH,phenol	C6H10,cyclo-	C6H12,1-hexene	C6H12,cyclo-	C6H13,n-hexyl
C6H14,n-hexane	C7H7,benzyl	C7H8	C7H8O,cresol-mx	C7H14,1-heptene
C7H15,n-heptyl	C7H16,n-heptane	C7H16,2-methylh	C8H8,styrene	C8H10,ethylbenz
C8H16,1-octene	C8H17,n-octyl	C8H18,n-octane	C8H18,isoctane	C9H19,n-nonyl
C10H8,naphthale	C10H21,n-decyl	C11H21	C12H9,o-bipheny	C12H10,biphenyl
HCN	HCO	HCCN	HCCO	HNC
HNCO	HNO	HNO2	HNO3	H02
HCHO,formaldehy	HCOOH	H2O2	(HCOOH)2	*N
NCO	*NH	NH2	NH3	NH2OH
NO2	NO3	N2O	NCN	N2H2
NH2NO2	N2H4	N2O3	N2O4	N2O5
N3	N3H	O3	THDCPD,endo	THDCPD,exo
N2H4(L)	C(gr)	CH3OH(L)	C2H5OH(L)	C6H14(L),n-hexa
C6H5NH2(L)	C6H6(L)	H2O(cr)	H2O(L)	

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

O/F = 1.400000

	EFFECTIVE FUEL h(2)/R	EFFECTIVE OXIDANT h(1)/R	MIXTURE h0/R
ENTHALPY (KG-MOL)(K)/KG	0.48930525E+02	0.00000000E+00	0.20387719E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.69330834E-02
*H	0.12896952E+00	0.00000000E+00	0.53737301E-01
*N	0.47845361E-01	0.21736532E-01	0.32615210E-01
*O	0.00000000E+00	0.43473063E-01	0.25359287E-01

POINT ITN	T	C	H	N	O
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Pinf/Pt = 1.754339

Pinf/Pt = 1.764418

Pinf/Pt = 1.764534

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA

CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 1.40000 %FUEL= 41.666667 R,EQ.RATIO= 1.606308 PHI,EQ.RATIO= 1.606308

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.7645	620.30
P, BAR	8.2737	4.6889	0.01334
T, K	2897.52	2676.43	855.95
RHO, KG/CU M	6.7511-1	4.1703-1	3.7402-3
H, KJ/KG	169.51	-497.65	-4506.69
U, KJ/KG	-1056.01	-1622.01	-4863.31
G, KJ/KG	-37460.3	-35256.2	-15622.9
S, KJ/(KG)(K)	12.9869	12.9869	12.9869

M, (1/n)	19.658	19.792	19.956
MW, MOL WT	19.658	19.792	19.956
(dLV/dLP)t	-1.00784	-1.00425	-1.00001
(dLV/dLT)p	1.1649	1.0954	1.0001
Cp, KJ/(KG)(K)	3.7157	3.1193	2.0133
GAMMAS	1.1718	1.1867	1.2610
SON VEL,M/SEC	1198.4	1155.1	670.6
MACH NUMBER	0.000	1.000	4.560

PERFORMANCE PARAMETERS

Ae/At	1.0000	42.115
CSTAR, M/SEC	1717.5	1717.5
CF	0.6726	1.7806

Ivac, M/SEC	2128.5	3174.8
Isp, M/SEC	1155.1	3058.2

MASS FRACTIONS

*CO	0.15702	0.15448	0.05009
*CO2	0.05841	0.06239	0.22642
*H	0.00089	0.00054	0.00000
*H2	0.01954	0.01967	0.02739
H2O	0.29698	0.30116	0.23927
*NO	0.00165	0.00068	0.00000
*N2	0.45606	0.45651	0.45683
*O	0.00041	0.00013	0.00000
*OH	0.00847	0.00425	0.00000
*O2	0.00056	0.00018	0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH4	CH3OH	CH3OOH	*CN
CNN	COOH	*C2	C2H	C2H2,acetylene
C2H2,vinylidene	CH2CO,ketene	O(CH)2O	HO(CO)2OH	C2H3,vinyl
CH3CN	CH3CO,acetyl	C2H4	C2H4O,ethylen-o	CH3CHO,ethanal
CH3COOH	OHCH2COOH	C2H5	C2H6	CH3N2CH3
C2H5OH	CH3OCH3	CH3O2CH3	CCN	CNC
OCCN	C2N2	C2O	*C3	C3H3,1-propynl
C3H3,2-propynl	C3H4,allene	C3H4,propyne	C3H4,cyclo-	C3H5,allyl
C3H6,propylene	C3H6,cyclo-	C3H6O,propylox	C3H6O,acetone	C3H6O,propanal
C3H7,n-propyl	C3H7,i-propyl	C3H8	C3H8O,1propanol	C3H8O,2propanol
CNCOCN	C3O2	*C4	C4H2,butadiyne	C4H4,1,3-cyclo-
C4H6,butadiene	C4H6,1butyne	C4H6,2butyne	C4H6,cyclo-	C4H8,1-butene
C4H8,cis2-buten	C4H8,tr2-butene	C4H8,isobutene	C4H8,cyclo-	(CH3COOH)2
C4H9,n-butyl	C4H9,i-butyl	C4H9,s-butyl	C4H9,t-butyl	C4H10,n-butane
C4H10,isobutane	C4N2	*C5	C5H6,1,3cyclo-	C5H8,cyclo-
C5H10,1-pentene	C5H10,cyclo-	C5H11,pentyl	C5H11,t-pentyl	C5H12,n-pentane
C5H12,i-pentane	CH3C(CH3)2CH3	C6H2	C6H5,phenyl	C6H5O,phenoxy
C6H6	C6H5OH,phenol	C6H10,cyclo-	C6H12,1-hexene	C6H12,cyclo-
C6H13,n-hexyl	C6H14,n-hexane	C7H7,benzyl	C7H8	C7H8O,cresol-mx
C7H14,1-heptene	C7H15,n-heptyl	C7H16,n-heptane	C7H16,2-methylh	C8H8,styrene
C8H10,ethylbenz	C8H16,1-octene	C8H17,n-octyl	C8H18,n-octane	C8H18,isooctane
C9H19,n-nonyl	C10H8,naphthale	C10H21,n-decyl	C11H21	C12H9,o-bipheny
C12H10,biphenyl	HCN	HCO	HCCN	HCCO
HNC	HNCO	HNO	HNO2	HNO3
HO2	HCHO,formaldehy	HCOOH	H2O2	(HCOOH)2
*N	NCO	*NH	NH2	NH3
NH2OH	NO2	NO3	N2O	NCN
N2H2	NH2NO2	N2H4	N2O3	N2O4
N2O5	N3	N3H	O3	THDCPD,endo
THDCPD,exo	N2H4(L)	C(gr)	CH3OH(L)	C2H5OH(L)
C6H14(L),n-hexa	C6H5NH2(L)	C6H6(L)	H2O(cr)	H2O(L)

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

O/F = 1.500000

ENTHALPY	EFFECTIVE FUEL	EFFECTIVE OXIDANT	MIXTURE
(KG-MOL)(K)/KG	h(2)/R	h(1)/R	h0/R
	0.48930525E+02	0.00000000E+00	0.19572210E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i

*C	0.16639400E-01	0.00000000E+00	0.66557601E-02
*H	0.12896952E+00	0.00000000E+00	0.51587809E-01
*N	0.47845361E-01	0.21736532E-01	0.32180063E-01
*O	0.00000000E+00	0.43473063E-01	0.26083838E-01

POINT	ITN	T	C	H	N	O
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Pinf/Pt = 1.746606
 Pinf/Pt = 1.754499
 Pinf/Pt = 1.754577

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA
 CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 1.50000 %FUEL= 40.000000 R,EQ.RATIO= 1.499221 PHI,EQ.RATIO= 1.499221

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.7546	620.30
P, BAR	8.2737	4.7155	0.01334
T, K	2972.59	2768.65	919.34
RHO, KG/CU M	6.7540-1	4.1675-1	3.5949-3
H, KJ/KG	162.73	-499.63	-4619.32
U, KJ/KG	-1062.27	-1631.11	-4990.35
G, KJ/KG	-37955.7	-36003.0	-16408.4
S, KJ/(KG)(K)	12.8233	12.8233	12.8233

M, (1/n)	20.176	20.345	20.602
MW, MOL WT	20.176	20.345	20.602
(dLV/dLP)t	-1.01110	-1.00658	-1.00000
(dLV/dLT)p	1.2311	1.1459	1.0000
Cp, KJ/(KG)(K)	4.2084	3.5199	1.9541
GAMMA _s	1.1592	1.1708	1.2603
SON VEL,M/SEC	1191.6	1151.0	683.8
MACH NUMBER	0.000	1.000	4.523

PERFORMANCE PARAMETERS

Ae/At	1.0000	43.145
CSTAR, M/SEC	1724.9	1724.9
CF	0.6673	1.7929
Ivac, M/SEC	2134.0	3212.6
Isp, M/SEC	1151.0	3092.6

MASS FRACTIONS

*CO	0.14459	0.14169	0.05040
*CO2	0.06573	0.07030	0.21373
*H	0.00101	0.00067	0.00000
*H2	0.01607	0.01603	0.02262
H2O	0.30523	0.31155	0.26251
*NO	0.00289	0.00141	0.00000
*N2	0.44938	0.45008	0.45074
*O	0.00085	0.00033	0.00000
*OH	0.01279	0.00736	0.00000

*O2 0.00144 0.00058 0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH4	CH3OH	CH3OOH	*CN
CNN	COOH	*C2	C2H	C2H2,acetylene
C2H2,vinylidene	CH2CO, ketene	O(CH)2O	HO(CO)2OH	C2H3,vinyl
CH3CN	CH3CO,acetyl	C2H4	C2H4O,ethylen-o	CH3CHO,ethanal
CH3COOH	OHCH2COOH	C2H5	C2H6	CH3N2CH3
C2H5OH	CH3OCH3	CH3O2CH3	CCN	CNC
OCCN	C2N2	C2O	*C3	C3H3,1-propynl
C3H3,2-propynl	C3H4,allene	C3H4,propyne	C3H4,cyclo-	C3H5,allyl
C3H6,propylene	C3H6,cyclo-	C3H6O,propylox	C3H6O,acetone	C3H6O,propanal
C3H7,n-propyl	C3H7,i-propyl	C3H8	C3H8O,1propanol	C3H8O,2propanol
CNCOCN	C3O2	*C4	C4H2,butadiyne	C4H4,1,3-cyclo-
C4H6,butadiene	C4H6,1butyne	C4H6,2butyne	C4H6,cyclo-	C4H8,1-butene
C4H8,cis2-buten	C4H8,tr2-butene	C4H8,isobutene	C4H8,cyclo-	(CH3COOH)2
C4H9,n-butyl	C4H9,i-butyl	C4H9,s-butyl	C4H9,t-butyl	C4H10,n-butane
C4H10,isobutane	C4N2	*C5	C5H6,1,3cyclo-	C5H8,cyclo-
C5H10,1-pentene	C5H10,cyclo-	C5H11,pentyl	C5H11,t-pentyl	C5H12,n-pentane
C5H12,i-pentane	CH3C(CH3)2CH3	C6H2	C6H5,phenyl	C6H5O,phenoxy
C6H6	C6H5OH,phenol	C6H10,cyclo-	C6H12,1-hexene	C6H12,cyclo-
C6H13,n-hexyl	C6H14,n-hexane	C7H7,benzyl	C7H8	C7H8O,cresol-mx
C7H14,1-heptene	C7H15,n-heptyl	C7H16,n-heptane	C7H16,2-methylh	C8H8,styrene
C8H10,ethylbenz	C8H16,1-octene	C8H17,n-octyl	C8H18,n-octane	C8H18,isooctane
C9H19,n-nonyl	C10H8,naphthale	C10H21,n-decyl	C11H21	C12H9,o-bipheny
C12H10,biphenyl	HNC	HCO	HCCN	HCCO
HNC	HNCO	HNO	HNO2	HNO3
HO2	HCHO,formaldehy	HCOOH	H2O2	(HCOOH)2
*N	NCO	*NH	NH2	NH3
NH2OH	NO2	NO3	N2O	NCN
N2H2	NH2NO2	N2H4	N2O3	N2O4
N2O5	N3	N3H	O3	THDCPD,endo
THDCPD,exo	N2H4(L)	C(gr)	CH3OH(L)	C2H5OH(L)
C6H14(L),n-hexa	C6H5NH2(L)	C6H6(L)	H2O(cr)	H2O(L)

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

O/F = 1.600000

ENTHALPY (KG-MOL)(K)/KG	EFFECTIVE FUEL h(2)/R	EFFECTIVE OXIDANT h(1)/R	MIXTURE h0/R
	0.48930525E+02	0.00000000E+00	0.18819433E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.63997693E-02
*H	0.12896952E+00	0.00000000E+00	0.49603662E-01
*N	0.47845361E-01	0.21736532E-01	0.31778389E-01
*O	0.00000000E+00	0.43473063E-01	0.26752654E-01

POINT ITN	T	C	H	N	O
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Pinf/Pt = 1.740376

Pinf/Pt = 1.745713

Pinf/Pt = 1.745752

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA

CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.500000	48900.000	298.150
FUEL	N2H4	0.500000	0.000	0.000
OXIDANT	N2O4	1.000000	0.000	0.000

O/F= 1.60000 %FUEL= 38.461538 R,EQ.RATIO= 1.405519 PHI,EQ.RATIO= 1.405519

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.7458	620.30
P, BAR	8.2737	4.7393	0.01334
T, K	3029.51	2841.84	984.51
RHO, KG/CU M	6.7846-1	4.1836-1	3.4602-3
H, KJ/KG	156.47	-498.74	-4716.75
U, KJ/KG	-1063.00	-1631.58	-5102.22
G, KJ/KG	-38218.3	-36496.3	-17187.6
S, KJ/(KG)(K)	12.6670	12.6670	12.6670
M, (1/n)	20.656	20.858	21.236
MW, MOL WT	20.656	20.858	21.236
(dLV/dLP)t	-1.01492	-1.00966	-1.00000
(dLV/dLT)p	1.3079	1.2120	1.0000
Cp, KJ/(KG)(K)	4.7627	4.0331	1.9049
GAMMAS	1.1490	1.1568	1.2587
SON VEL,M/SEC	1183.7	1144.7	696.6
MACH NUMBER	0.000	1.000	4.482

PERFORMANCE PARAMETERS

Ae/At	1.0000	44.333
CSTAR, M/SEC	1727.6	1727.6
CF	0.6626	1.8071
Ivac, M/SEC	2134.3	3245.4
Isp, M/SEC	1144.7	3121.9

MASS FRACTIONS

*CO	0.13255	0.12901	0.04917
*CO2	0.07339	0.07895	0.20440
*H	0.00108	0.00077	0.00000
H02	0.00001	0.00000	0.00000
*H2	0.01319	0.01295	0.01833
H2O	0.30996	0.31819	0.28299
*NO	0.00456	0.00255	0.00000
*N2	0.44298	0.44392	0.44511
*O	0.00152	0.00072	0.00000
*OH	0.01760	0.01134	0.00000
*O2	0.00316	0.00159	0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH4	CH3OH	CH3OOH	*CN
CNN	COOH	*C2	C2H	C2H2,acetylene
C2H2,vinylidene	CH2CO,ketene	O(CH)2O	HO(CO)2OH	C2H3,vinyl
CH3CN	CH3CO,acetyl	C2H4	C2H4O,ethylen-o	CH3CHO,ethanal
CH3COOH	OHCH2COOH	C2H5	C2H6	CH3N2CH3
C2H5OH	CH3OCH3	CH3O2CH3	CCN	CNC

OCCN	C2N2	C2O	*C3	C3H3,1-propyn1
C3H3,2-propyn1	C3H4,allene	C3H4,propyne	C3H4,cyclo-	C3H5,allyl
C3H6,propylene	C3H6,cyclo-	C3H6O,propylox	C3H6O,acetone	C3H6O,propanal
C3H7,n-propyl	C3H7,i-propyl	C3H8	C3H8O,1propanol	C3H8O,2propanol
CNCOCN	C3O2	*C4	C4H2,butadiyne	C4H4,1,3-cyclo-
C4H6,butadiene	C4H6,1butyne	C4H6,2butyne	C4H6,cyclo-	C4H8,1-butene
C4H8,cis2-buten	C4H8,tr2-butene	C4H8,isobutene	C4H8,cyclo-	(CH3COOH)2
C4H9,n-butyl	C4H9,i-butyl	C4H9,s-butyl	C4H9,t-butyl	C4H10,n-butane
C4H10,isobutane	C4N2	*C5	C5H6,1,3cyclo-	C5H8,cyclo-
C5H10,1-pentene	C5H10,cyclo-	C5H11,pentyl	C5H11,t-pentyl	C5H12,n-pentane
C5H12,i-pentane	CH3C(CH3)2CH3	C6H2	C6H5,phenyl	C6H5O,phenoxy
C6H6	C6H5OH,phenol	C6H10,cyclo-	C6H12,1-hexene	C6H12,cyclo-
C6H13,n-hexyl	C6H14,n-hexane	C7H7,benzyl	C7H8	C7H8O,cresol-mx
C7H14,1-heptene	C7H15,n-heptyl	C7H16,n-heptane	C7H16,2-methylh	C8H8,styrene
C8H10,ethylbenz	C8H16,1-octene	C8H17,n-octyl	C8H18,n-octane	C8H18,isooctane
C9H19,n-nonyl	C10H8,naphthale	C10H21,n-decyl	C11H21	C12H9,o-bipheny
C12H10,biphenyl	HCN	HCO	HCCN	HCCO
HNC	HNCO	HNO	HNO2	HNO3
HCHO,formaldehy	HC00H	H2O2	(HC00H)2	*N
NCO	*NH	NH2	NH3	NH2OH
NO2	NO3	N2O	NCN	N2H2
NH2NO2	N2H4	N2O3	N2O4	N2O5
N3	N3H	O3	THDCPD,endo	THDCPD,exo
N2H4(L)	C(gr)	CH3OH(L)	C2H5OH(L)	C6H14(L),n-hexa
C6H5NH2(L)	C6H6(L)	H2O(cr)	H2O(L)	

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

$$O/F = 1.700000$$

ENTHALPY (KG-MOL)(K)/KG	EFFECTIVE FUEL h(2)/R 0.48930525E+02	EFFECTIVE OXIDANT h(1)/R 0.00000000E+00	MIXTURE h0/R 0.18122417E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.61627408E-02
*H	0.12896952E+00	0.00000000E+00	0.47766490E-01
*N	0.47845361E-01	0.21736532E-01	0.31406468E-01
*O	0.00000000E+00	0.43473063E-01	0.27371929E-01

POINT ITN T C H N O

$$P_{inf}/P_t = 1.735703$$

$$P_{inf}/P_t = 1.738594$$

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

$$P_{in} = 120.0 \text{ PSIA}$$

$$CASE = kaw00$$

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

$$O/F = 1.70000 \quad \%FUEL = 37.037037 \quad R, EQ. RATIO = 1.322842 \quad PHI, EQ. RATIO = 1.322842$$

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.7386	620.30
P, BAR	8.2737	4.7588	0.01334
T, K	3070.31	2896.30	1051.69
RHO, KG/CU M	6.8378-1	4.2149-1	3.3342-3
H, KJ/KG	150.68	-495.99	-4799.87
U, KJ/KG	-1059.31	-1625.04	-5199.92
G, KJ/KG	-38283.2	-36751.6	-17964.8
S, KJ/(KG)(K)	12.5179	12.5179	12.5179
M, (1/n)	21.098	21.329	21.858
MW, MOL WT	21.098	21.329	21.858
(dLV/dLP)t	-1.01895	-1.01338	-1.00000
(dLV/dLT)p	1.3883	1.2910	1.0000
Cp, KJ/(KG)(K)	5.3196	4.6269	1.8642
GAMMAS	1.1414	1.1455	1.2564
SON VEL,M/SEC	1175.2	1137.3	708.9
MACH NUMBER	0.000	1.000	4.438

PERFORMANCE PARAMETERS

Ae/At	1.0000	45.689
CSTAR, M/SEC	1726.1	1726.1
CF	0.6589	1.8230
Ivac, M/SEC	2130.0	3273.7
Isp, M/SEC	1137.3	3146.6

MASS FRACTIONS

*CO	0.12102	0.11666	0.04640
*CO2	0.08107	0.08792	0.19831
*H	0.00109	0.00082	0.00000
H02	0.00001	0.00001	0.00000
*H2	0.01084	0.01044	0.01447
H2O	0.31168	0.32130	0.30091
*NO	0.00656	0.00412	0.00000
*N2	0.43683	0.43797	0.43990
*O	0.00238	0.00132	0.00000
*OH	0.02246	0.01580	0.00000
*O2	0.00603	0.00363	0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH4	CH3OH	CH3OOH	*CN
CNN	COOH	*C2	C2H	C2H2,acetylene
C2H2,vinylidene	CH2CO,ketene	O(CH)2O	HO(CO)2OH	C2H3,vinyl
CH3CN	CH3CO,acetyl	C2H4	C2H4O,ethylen-o	CH3CHO,ethanal
CH3COOH	OHCH2COOH	C2H5	C2H6	CH3N2CH3
C2H5OH	CH3OCH3	CH3O2CH3	CCN	CNC
OCCN	C2N2	C2O	*C3	C3H3,1-propynl
C3H3,2-propynl	C3H4,allene	C3H4,propyne	C3H4,cyclo-	C3H5,allyl
C3H6,propylene	C3H6,cyclo-	C3H6O,propylox	C3H6O,acetone	C3H6O,propanal
C3H7,n-propyl	C3H7,i-propyl	C3H8	C3H8O,1propanol	C3H8O,2propanol
CNCOCN	C3O2	*C4	C4H2,butadiyne	C4H4,1,3-cyclo-
C4H6,butadiene	C4H6,1butyne	C4H6,2butyne	C4H6,cyclo-	C4H8,1-butene
C4H8,cis2-buten	C4H8,tr2-butene	C4H8,isobutene	C4H8,cyclo-	(CH3COOH)2
C4H9,n-butyl	C4H9,i-butyl	C4H9,s-butyl	C4H9,t-butyl	C4H10,n-butane
C4H10,isobutane	C4N2	*C5	C5H6,1,3cyclo-	C5H8,cyclo-
C5H10,1-pentene	C5H10,cyclo-	C5H11,pentyl	C5H11,t-pentyl	C5H12,n-pentane
C5H12,i-pentane	CH3C(CH3)2CH3	C6H2	C6H5,phenyl	C6H5O,phenoxy
C6H6	C6H5OH,phenol	C6H10,cyclo-	C6H12,1-hexene	C6H12,cyclo-
C6H13,n-hexyl	C6H14,n-hexane	C7H7,benzyl	C7H8	C7H8O,cresol-mx
C7H14,1-heptene	C7H15,n-heptyl	C7H16,n-heptane	C7H16,2-methylh	C8H8,styrene
C8H10,ethylbenz	C8H16,1-octene	C8H17,n-octyl	C8H18,n-octane	C8H18,isooctane

C9H19,n-nonyl	C10H8,naphthale	C10H21,n-decyl	C11H21	C12H9,o-bipheny
C12H10,biphenyl	HCN	HCO	HCCN	HCCO
HNC	HNCO	HNO	HNO2	HN03
HCHO,formaldehy	HCOOH	H2O2	(HCOOH)2	*N
NCO	*NH	NH2	NH3	NH2OH
NO2	NO3	N2O	NCN	N2H2
NH2NO2	N2H4	N2O3	N2O4	N2O5
N3	N3H	O3	THDCPD,endo	THDCPD,exo
N2H4(L)	C(gr)	CH3OH(L)	C2H5OH(L)	C6H14(L),n-hexa
C6H5NH2(L)	C6H6(L)	H2O(cr)	H2O(L)	

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

O/F = 1.800000

ENTHALPY (KG-MOL)(K)/KG	EFFECTIVE FUEL h(2)/R	EFFECTIVE OXIDANT h(1)/R	MIXTURE h0/R
	0.48930525E+02	0.00000000E+00	0.17475187E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.59426429E-02
*H	0.12896952E+00	0.00000000E+00	0.46060544E-01
*N	0.47845361E-01	0.21736532E-01	0.31061113E-01
*O	0.00000000E+00	0.43473063E-01	0.27946969E-01

POINT ITN	T	C	H	N	O
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Pinf/Pt = 1.732430

Pinf/Pt = 1.733400

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA

CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 1.80000 %FUEL= 35.714286 R,EQ.RATIO= 1.249350 PHI,EQ.RATIO= 1.249350

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.7334	620.30
P, BAR	8.2737	4.7731	0.01334
T, K	3097.55	2933.77	1121.23
RHO, KG/CU M	6.9084-1	4.2576-1	3.2150-3
H, KJ/KG	145.30	-492.24	-4869.19
U, KJ/KG	-1052.32	-1613.31	-5284.08
G, KJ/KG	-38189.8	-36800.4	-18745.4
S, KJ/(KG)(K)	12.3759	12.3759	12.3759
M, (1/n)	21.505	21.759	22.470
MW, MOL WT	21.505	21.759	22.470
(dLV/dLP)t	-1.02267	-1.01727	-1.00000
(dLV/dLT)p	1.4621	1.3728	1.0000
Cp, KJ/(KG)(K)	5.8044	5.2168	1.8312

GAMMAS	1.1360	1.1374	1.2532
SON VEL,M/SEC	1166.4	1129.2	721.1
MACH NUMBER	0.000	1.000	4.392

PERFORMANCE PARAMETERS

Ae/At	1.0000	47.221
CSTAR, M/SEC	1720.9	1720.9
CF	0.6562	1.8402
Ivac, M/SEC	2122.0	3297.9
Isp, M/SEC	1129.2	3166.9

MASS FRACTIONS

*CO	0.11014	0.10489	0.04209
*CO2	0.08848	0.09672	0.19540
*H	0.00107	0.00082	0.00000
H02	0.00002	0.00001	0.00000
*H2	0.00896	0.00844	0.01102
H2O	0.31102	0.32142	0.31643
*NO	0.00877	0.00600	0.00000
*N2	0.43096	0.43226	0.43506
*O	0.00336	0.00210	0.00000
*OH	0.02694	0.02022	0.00000
*O2	0.01026	0.00710	0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH4	CH3OH	CH3OOH	*CN
CNN	COOH	*C2	C2H	C2H2,acetylene
C2H2,vinylidene	CH2CO,ketene	O(CH)2O	HO(CO)2OH	C2H3,vinyl
CH3CN	CH3CO,acetyl	C2H4	C2H4O,ethylen-o	CH3CHO,ethanal
CH3COOH	OHCH2COOH	C2H5	C2H6	CH3N2CH3
C2H5OH	CH3OCH3	CH3O2CH3	CCN	CNC
OCCN	C2N2	C2O	*C3	C3H3,1-propyn1
C3H3,2-propyn1	C3H4,allene	C3H4,propyne	C3H4,cyclo-	C3H5,allyl
C3H6,propylene	C3H6,cyclo-	C3H6O,propylox	C3H6O,acetone	C3H6O,propanal
C3H7,n-propyl	C3H7,i-propyl	C3H8	C3H8O,1propanol	C3H8O,2propanol
CNCOCN	C3O2	*C4	C4H2,butadiyne	C4H4,1,3-cyclo-
C4H6,butadiene	C4H6,1butyne	C4H6,2butyne	C4H6,cyclo-	C4H8,1-butene
C4H8,cis2-buten	C4H8,tr2-butene	C4H8,isobutene	C4H8,cyclo-	(CH3COOH)2
C4H9,n-butyl	C4H9,i-butyl	C4H9,s-butyl	C4H9,t-butyl	C4H10,n-butane
C4H10,isobutane	C4N2	*C5	C5H6,1,3cyclo-	C5H8,cyclo-
C5H10,1-pentene	C5H10,cyclo-	C5H11,pentyl	C5H11,t-pentyl	C5H12,n-pentane
C5H12,i-pentane	CH3C(CH3)2CH3	C6H2	C6H5,phenyl	C6H5O,phenoxy
C6H6	C6H5OH,phenol	C6H10,cyclo-	C6H12,1-hexene	C6H12,cyclo-
C6H13,n-hexyl	C6H14,n-hexane	C7H7,benzyl	C7H8	C7H8O,cresol-mx
C7H14,1-heptene	C7H15,n-heptyl	C7H16,n-heptane	C7H16,2-methylh	C8H8,styrene
C8H10,ethylbenz	C8H16,1-octene	C8H17,n-octyl	C8H18,n-octane	C8H18,isooctane
C9H19,n-nonyl	C10H8,naphthale	C10H21,n-decyl	C11H21	C12H9,o-bipheny
C12H10,biphenyl	HCN	HCO	HCCN	HCCO
HNC	HNCO	HNO	HNO2	HNO3
HCHO,formaldehy	HC00H	H2O2	(HC00H)2	*N
NCO	*NH	NH2	NH3	NH2OH
NO2	NO3	N2O	NCN	N2H2
NH2NO2	N2H4	N2O3	N2O4	N2O5
N3	N3H	O3	THDCPD,endo	THDCPD,exo
N2H4(L)	C(gr)	CH3OH(L)	C2H5OH(L)	C6H14(L),n-hexa
C6H5NH2(L)	C6H6(L)	H2O(cr)	H2O(L)	

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

O/F = 1.900000

ENTHALPY (KG-MOL)(K)/KG	EFFECTIVE FUEL h(2)/R	EFFECTIVE OXIDANT h(1)/R	MIXTURE h0/R
	0.48930525E+02	0.00000000E+00	0.16872595E+02

KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.57377242E-02
*H	0.12896952E+00	0.00000000E+00	0.44472249E-01
*N	0.47845361E-01	0.21736532E-01	0.30739576E-01
*O	0.00000000E+00	0.43473063E-01	0.28482352E-01

POINT ITN	T	C	H	N	O
Pinf/Pt = 1.730269					
Pinf/Pt = 1.729966					

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA
CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 1.90000 %FUEL= 34.482759 R,EQ.RATIO= 1.183595 PHI,EQ.RATIO= 1.183595

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.7300	620.30
P, BAR	8.2737	4.7826	0.01334
T, K	3113.95	2957.11	1193.63
RHO, KG/CU M	6.9919-1	4.3085-1	3.1007-3
H, KJ/KG	140.29	-487.98	-4924.84
U, KJ/KG	-1043.04	-1598.00	-5355.00
G, KJ/KG	-37977.2	-36685.5	-19535.9
S, KJ/(KG)(K)	12.2409	12.2409	12.2409

M, (1/n)	21.880	22.150	23.071
MW, MOL WT	21.880	22.150	23.071
(dLV/dLP)t	-1.02563	-1.02068	-1.00000
(dLV/dLT)p	1.5205	1.4440	1.0000
Cp, KJ/(KG)(K)	6.1605	5.7012	1.8048
GAMMA _s	1.1325	1.1320	1.2495
SON VEL,M/SEC	1157.6	1121.0	733.1
MACH NUMBER	0.000	1.000	4.341

PERFORMANCE PARAMETERS

Ae/At	1.0000	48.937
CSTAR, M/SEC	1713.1	1713.1
CF	0.6543	1.8579
Ivac, M/SEC	2111.2	3318.0
Isp, M/SEC	1121.0	3182.8

MASS FRACTIONS

*CO	0.10000	0.09393	0.03618
*CO2	0.09538	0.10492	0.19566
*H	0.00101	0.00079	0.00000
H02	0.00003	0.00002	0.00000
*H2	0.00746	0.00687	0.00794
H2O	0.30859	0.31929	0.32966
*NO	0.01105	0.00805	0.00000
NO2	0.00001	0.00000	0.00000
*N2	0.42539	0.42680	0.43056
*O	0.00437	0.00296	0.00000
*OH	0.03079	0.02417	0.00000
*O2	0.01589	0.01219	0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH4	CH3OH	CH3OOH	*CN
CNN	COOH	*C2	C2H	C2H2,acetylene
C2H2,vinylidene	CH2CO,ketene	O(CH)2O	HO(CO)2OH	C2H3,vinyl
CH3CN	CH3CO,acetyl	C2H4	C2H4O,ethylen-o	CH3CHO,ethanal
CH3COOH	OHCH2COOH	C2H5	C2H6	CH3N2CH3
C2H5OH	CH3OCH3	CH3O2CH3	CCN	CNC
OCCN	C2N2	C2O	*C3	C3H3,1-propynl
C3H3,2-propynl	C3H4,allene	C3H4,propyne	C3H4,cyclo-	C3H5,allyl
C3H6,propylene	C3H6,cyclo-	C3H6O,propylox	C3H6O,acetone	C3H6O,propanal
C3H7,n-propyl	C3H7,i-propyl	C3H8	C3H8O,1propanol	C3H8O,2propanol
CNCOCN	C3O2	*C4	C4H2,butadiyne	C4H4,1,3-cyclo-
C4H6,butadiene	C4H6,1butyne	C4H6,2butyne	C4H6,cyclo-	C4H8,1-butene
C4H8,cis2-buten	C4H8,tr2-butene	C4H8,isobutene	C4H8,cyclo-	(CH3COOH)2
C4H9,n-butyl	C4H9,i-butyl	C4H9,s-butyl	C4H9,t-butyl	C4H10,n-butane
C4H10,isobutane	C4N2	*C5	C5H6,1,3cyclo-	C5H8,cyclo-
C5H10,1-pentene	C5H10,cyclo-	C5H11,pentyl	C5H11,t-pentyl	C5H12,n-pentane
C5H12,i-pentane	CH3C(CH3)2CH3	C6H2	C6H5,phenyl	C6H5O,phenoxy
C6H6	C6H5OH,phenol	C6H10,cyclo-	C6H12,1-hexene	C6H12,cyclo-
C6H13,n-hexyl	C6H14,n-hexane	C7H7,benzyl	C7H8	C7H8O,cresol-mx
C7H14,1-heptene	C7H15,n-heptyl	C7H16,n-heptane	C7H16,2-methylh	C8H8,styrene
C8H10,ethylbenz	C8H16,1-octene	C8H17,n-octyl	C8H18,n-octane	C8H18,isoctane
C9H19,n-nonyl	C10H8,naphthale	C10H21,n-decyl	C11H21	C12H9,o-bipheny
C12H10,biphenyl	HCN	HCO	HCCN	HCCO
HNC	HNCO	HNO	HN02	HN03
HCHO,formaldehy	HCOOH	H2O2	(HCOOH)2	*N
NCO	*NH	NH2	NH3	NH2OH
NO3	N2O	NCN	N2H2	NH2NO2
N2H4	N2O3	N2O4	N2O5	N3
N3H	O3	THDCPD,endo	THDCPD,exo	N2H4(L)
C(gr)	CH3OH(L)	C2H5OH(L)	C6H14(L),n-hexa	C6H5NH2(L)
C6H6(L)	H2O(cr)	H2O(L)		

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

O/F = 2.000000

ENTHALPY (KG-MOL)(K)/KG	EFFECTIVE FUEL h(2)/R 0.48930525E+02	EFFECTIVE OXIDANT h(1)/R 0.00000000E+00	MIXTURE h0/R 0.16310175E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.55464667E-02
*H	0.12896952E+00	0.00000000E+00	0.42989841E-01
*N	0.47845361E-01	0.21736532E-01	0.30439475E-01
*O	0.00000000E+00	0.43473063E-01	0.28982042E-01

POINT ITN T C H N O

Pinf/Pt = 1.728914

Pinf/Pt = 1.727880

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA

CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 2.00000 %FUEL= 33.333333 R,EQ.RATIO= 1.124415 PHI,EQ.RATIO= 1.124415

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.7279	620.30
P, BAR	8.2737	4.7883	0.01334
T, K	3121.99	2969.46	1269.72
RHO, KG/CU M	7.0843-1	4.3651-1	2.9896-3
H, KJ/KG	135.61	-483.47	-4966.35
U, KJ/KG	-1032.28	-1580.43	-5412.51
G, KJ/KG	-37679.3	-36450.8	-20345.8
S, KJ/(KG)(K)	12.1124	12.1124	12.1124

M, (1/n)	22.226	22.507	23.662
MW, MOL WT	22.226	22.507	23.662
(dLV/dLP)t	-1.02761	-1.02312	-1.00000
(dLV/dLT)p	1.5597	1.4948	1.0000
Cp, KJ/(KG)(K)	6.3697	6.0175	1.7841
GAMMAS	1.1303	1.1287	1.2453
SON VEL,M/SEC	1148.9	1112.7	745.4
MACH NUMBER	0.000	1.000	4.286

PERFORMANCE PARAMETERS

Ae/At	1.0000	50.861
CSTAR, M/SEC	1703.4	1703.4
CF	0.6532	1.8753
Ivac, M/SEC	2098.6	3334.0
Isp, M/SEC	1112.7	3194.4

MASS FRACTIONS

*CO	0.09066	0.08391	0.02852
*CO2	0.10164	0.11226	0.19928
*H	0.00094	0.00074	0.00000
H02	0.00004	0.00002	0.00000
*H2	0.00626	0.00565	0.00522
H2O	0.30495	0.31559	0.34062
*NO	0.01328	0.01010	0.00000
NO2	0.00001	0.00001	0.00000
*N2	0.42015	0.42163	0.42636
*O	0.00533	0.00381	0.00000
*OH	0.03391	0.02741	0.00000
*O2	0.02282	0.01886	0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH4	CH3OH	CH3OOH	*CN
CNN	COOH	*C2	C2H	C2H2, acetylene
C2H2, vinylidene	CH2CO, ketene	O(CH)2O	HO(CO)2OH	C2H3, vinyl
CH3CN	CH3CO, acetyl	C2H4	C2H4O, ethylene-o	CH3CHO, ethanal
CH3COOH	OHCH2COOH	C2H5	C2H6	CH3N2CH3
C2H5OH	CH3OCH3	CH3O2CH3	CCN	CNC
OCCN	C2N2	C2O	*C3	C3H3, 1-propynl
C3H3, 2-propynl	C3H4, allene	C3H4, propyne	C3H4, cyclo-	C3H5, allyl
C3H6, propylene	C3H6, cyclo-	C3H6O, propylox	C3H6O, acetone	C3H6O, propanal
C3H7, n-propyl	C3H7, i-propyl	C3H8	C3H8O, 1propanol	C3H8O, 2propanol
CNCOCN	C3O2	*C4	C4H2, butadiyne	C4H4, 1,3-cyclo-
C4H6, butadiene	C4H6, 1butyne	C4H6, 2butyne	C4H6, cyclo-	C4H8, 1-butene
C4H8, cis2-buten	C4H8, tr2-butene	C4H8, isobutene	C4H8, cyclo-	(CH3COOH)2
C4H9, n-butyl	C4H9, i-butyl	C4H9, s-butyl	C4H9, t-butyl	C4H10, n-butane
C4H10, isobutane	C4N2	*C5	C5H6, 1,3cyclo-	C5H8, cyclo-
C5H10, 1-pentene	C5H10, cyclo-	C5H11, pentyl	C5H11, t-pentyl	C5H12, n-pentane
C5H12, i-pentane	CH3C(CH3)2CH3	C6H2	C6H5, phenyl	C6H5O, phenoxy
C6H6	C6H5OH, phenol	C6H10, cyclo-	C6H12, 1-hexene	C6H12, cyclo-
C6H13, n-hexyl	C6H14, n-hexane	C7H7, benzyl	C7H8	C7H8O, cresol-mx
C7H14, 1-heptene	C7H15, n-heptyl	C7H16, n-heptane	C7H16, 2-methylh	C8H8, styrene
C8H10, ethylbenz	C8H16, 1-octene	C8H17, n-octyl	C8H18, n-octane	C8H18, isooctane
C9H19, n-nonyl	C10H8, naphthale	C10H21, n-decyl	C11H21	C12H9, o-bipheny
C12H10, biphenyl	HCN	HCO	HCCN	HCCO
HNC	HNCO	HNO	HNO2	HNO3
HCHO, formaldehy	HCOOH	H2O2	(HCOOH)2	*N
NCO	*NH	NH2	NH3	NH2OH
NO3	N2O	NCN	N2H2	NH2NO2
N2H4	N2O3	N2O4	N2O5	N3
N3H	O3	THDCPD, endo	THDCPD, exo	N2H4(L)
C(gr)	CH3OH(L)	C2H5OH(L)	C6H14(L), n-hexa	C6H5NH2(L)
C6H6(L)	H2O(cr)	H2O(L)		

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

$$O/F = 2.100000$$

ENTHALPY (KG-MOL)(K)/KG	EFFECTIVE FUEL h(2)/R 0.48930525E+02	EFFECTIVE OXIDANT h(1)/R 0.00000000E+00	MIXTURE h0/R 0.15784040E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.53675485E-02
*H	0.12896952E+00	0.00000000E+00	0.41603072E-01
*N	0.47845361E-01	0.21736532E-01	0.30158735E-01
*O	0.00000000E+00	0.43473063E-01	0.29449494E-01

POINT ITN	T	C	H	N	O
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$$P_{inf}/P_t = 1.728115$$

$$P_{inf}/P_t = 1.726722$$

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA
CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 2.10000 %FUEL= 32.258065 R,EQ.RATIO= 1.070872 PHI,EQ.RATIO= 1.070872

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.7267	620.30
P, BAR	8.2737	4.7916	0.01334
T, K	3123.71	2973.64	1350.94
RHO, KG/CU M	7.1827-1	4.4255-1	2.8788-3
H, KJ/KG	131.24	-478.82	-4992.14
U, KJ/KG	-1020.65	-1561.54	-5455.46
G, KJ/KG	-37323.0	-36133.7	-21190.3
S, KJ/(KG)(K)	11.9903	11.9903	11.9903
M, (1/n)	22.547	22.835	24.243
MW, MOL WT	22.547	22.835	24.243
(dLV/dLP)t	-1.02863	-1.02446	-1.00000
(dLV/dLT)p	1.5804	1.5231	1.0001
Cp, KJ/(KG)(K)	6.4464	6.1618	1.7685
GAMMAS	1.1290	1.1269	1.2406
SON VEL,M/SEC	1140.4	1104.6	758.2
MACH NUMBER	0.000	1.000	4.222

PERFORMANCE PARAMETERS

Ae/At	1.0000	53.045
CSTAR, M/SEC	1692.5	1692.5
CF	0.6526	1.8913
Ivac, M/SEC	2084.8	3345.8
Isp, M/SEC	1104.6	3201.1

MASS FRACTIONS

*CO	0.08211	0.07484	0.01888
*CO2	0.10720	0.11863	0.20656
*H	0.00086	0.00068	0.00000
HNO	0.00001	0.00000	0.00000
HO2	0.00005	0.00003	0.00000
*H2	0.00529	0.00469	0.00285
H2O	0.30050	0.31090	0.34929
H2O2	0.00001	0.00000	0.00000
*NO	0.01538	0.01206	0.00000
NO2	0.00001	0.00001	0.00000
*N2	0.41523	0.41679	0.42242
*O	0.00619	0.00458	0.00000
*OH	0.03629	0.02989	0.00000
*O2	0.03085	0.02689	0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH4	CH3OH	CH3OOH	*CN
CNN	COOH	*C2	C2H	C2H2,acetylene
C2H2,vinylidene	CH2CO,ketene	O(CH)2O	HO(CO)2OH	C2H3,vinyl
CH3CN	CH3CO,acetyl	C2H4	C2H4O,ethylen-o	CH3CHO,ethanal
CH3COOH	OHCH2COOH	C2H5	C2H6	CH3N2CH3
C2H5OH	CH3OCH3	CH3O2CH3	CCN	CNC
OCCN	C2N2	C2O	*C3	C3H3,1-propynl

C3H3,2-propyn1	C3H4,allene	C3H4,propyne	C3H4,cyclo-	C3H5,allyl
C3H6,propylene	C3H6,cyclo-	C3H60,propylox	C3H60,acetone	C3H60,propanal
C3H7,n-propyl	C3H7,i-propyl	C3H8	C3H80,1propanol	C3H80,2propanol
CNCOCN	C3O2	*C4	C4H2,butadiyne	C4H4,1,3-cyclo-
C4H6,butadiene	C4H6,1butyne	C4H6,2butyne	C4H6,cyclo-	C4H8,1-butene
C4H8,cis2-buten	C4H8,tr2-butene	C4H8,isobutene	C4H8,cyclo-	(CH3COOH)2
C4H9,n-butyl	C4H9,i-butyl	C4H9,s-butyl	C4H9,t-butyl	C4H10,n-butane
C4H10,isobutane	C4N2	*C5	C5H6,1,3cyclo-	C5H8,cyclo-
C5H10,1-pentene	C5H10,cyclo-	C5H11,pentyl	C5H11,t-pentyl	C5H12,n-pentane
C5H12,i-pentane	CH3C(CH3)2CH3	C6H2	C6H5,phenyl	C6H5O,phenoxy
C6H6	C6H5OH,phenol	C6H10,cyclo-	C6H12,1-hexene	C6H12,cyclo-
C6H13,n-hexyl	C6H14,n-hexane	C7H7,benzyl	C7H8	C7H8O,cresol-mx
C7H14,1-heptene	C7H15,n-heptyl	C7H16,n-heptane	C7H16,2-methylh	C8H8,styrene
C8H10,ethylbenz	C8H16,1-octene	C8H17,n-octyl	C8H18,n-octane	C8H18,isoctane
C9H19,n-nonyl	C10H8,naphthale	C10H21,n-decyl	C11H21	C12H9,o-bipheny
C12H10,biphenyl	HCN	HCO	HCCN	HCCO
HNC	HNCO	HNO2	HNO3	HCHO,formaldehy
HCOOH	(HCOOH)2	*N	NCO	*NH
NH2	NH3	NH2OH	NO3	N2O
NCN	N2H2	NH2NO2	N2H4	N2O3
N2O4	N2O5	N3	N3H	O3
THDCPD,endo	THDCPD,exo	N2H4(L)	C(gr)	CH3OH(L)
C2H5OH(L)	C6H14(L),n-hexa	C6H5NH2(L)	C6H6(L)	H2O(cr)
H2O(L)				

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

O/F = 2.200000

ENTHALPY (KG-MOL)(K)/KG	EFFECTIVE FUEL h(2)/R 0.48930525E+02	EFFECTIVE OXIDANT h(1)/R 0.00000000E+00	MIXTURE h0/R 0.15290789E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.51998126E-02
*H	0.12896952E+00	0.00000000E+00	0.40302976E-01
*N	0.47845361E-01	0.21736532E-01	0.29895541E-01
*O	0.00000000E+00	0.43473063E-01	0.29887731E-01

POINT ITN	T	C	H	N	O
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Pinf/Pt = 1.727692

Pinf/Pt = 1.726172

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA

CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 2.20000 %FUEL= 31.250000 R,EQ.RATIO= 1.022196 PHI,EQ.RATIO= 1.022196

CHAMBER	THROAT	EXIT
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Pinf/P	1.0000	1.7262	620.30
P, BAR	8.2737	4.7931	0.01334
T, K	3120.69	2971.85	1440.70
RHO, KG/CU M	7.2851-1	4.4883-1	2.7631-3
H, KJ/KG	127.14	-474.11	-4997.05
U, KJ/KG	-1008.57	-1542.01	-5479.79
G, KJ/KG	-36928.4	-35762.3	-22104.2
S, KJ/(KG)(K)	11.8742	11.8742	11.8742

M, (1/n)	22.847	23.138	24.814
MW, MOL WT	22.847	23.138	24.814
(dLV/dLP)t	-1.02885	-1.02486	-1.00000
(dLV/dLT)p	1.5858	1.5321	1.0002
Cp, KJ/(KG)(K)	6.4205	6.1670	1.7598
GAMMAS	1.1283	1.1260	1.2353
SON VEL,M/SEC	1132.0	1096.6	772.2
MACH NUMBER	0.000	1.000	4.146

PERFORMANCE PARAMETERS

Ae/At	1.0000	55.642
CSTAR, M/SEC	1681.0	1681.0
CF	0.6523	1.9044
Ivac, M/SEC	2070.4	3352.1
Isp, M/SEC	1096.6	3201.3

MASS FRACTIONS

*CO	0.07433	0.06670	0.00688
*CO2	0.11205	0.12404	0.21803
*H	0.00078	0.00061	0.00000
HNO	0.00001	0.00000	0.00000
HO2	0.00006	0.00004	0.00000
*H2	0.00451	0.00393	0.00084
H2O	0.29557	0.30561	0.35550
H2O2	0.00001	0.00000	0.00000
*NO	0.01731	0.01385	0.00000
NO2	0.00002	0.00001	0.00000
*N2	0.41064	0.41226	0.41874
*O	0.00691	0.00523	0.00000
*OH	0.03800	0.03166	0.00001
*O2	0.03979	0.03604	0.00000

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH4	CH3OH	CH3OOH	*CN
CNN	COOH	*C2	C2H	C2H2,acetylene
C2H2,vinylidene	CH2CO,ketene	O(CH)2O	HO(CO)2OH	C2H3,vinyl
CH3CN	CH3CO,acetyl	C2H4	C2H4O,ethylen-o	CH3CHO,ethanal
CH3COOH	OHCH2COOH	C2H5	C2H6	CH3N2CH3
C2H5OH	CH3OCH3	CH3O2CH3	CCN	CNC
OCCN	C2N2	C2O	*C3	C3H3,1-propyn1
C3H3,2-propyn1	C3H4,allene	C3H4,propyne	C3H4,cyclo-	C3H5,allyl
C3H6,propylene	C3H6,cyclo-	C3H6O,propylox	C3H6O,acetone	C3H6O,propanal
C3H7,n-propyl	C3H7,i-propyl	C3H8	C3H8O,1propanol	C3H8O,2propanol
CNCOCN	C3O2	*C4	C4H2,butadiyne	C4H4,1,3-cyclo-
C4H6,butadiene	C4H6,1butyne	C4H6,2butyne	C4H6,cyclo-	C4H8,1-butene
C4H8,cis2-buten	C4H8,tr2-butene	C4H8,isobutene	C4H8,cyclo-	(CH3COOH)2
C4H9,n-butyl	C4H9,i-butyl	C4H9,s-butyl	C4H9,t-butyl	C4H10,n-butane
C4H10,isobutane	C4N2	*C5	C5H6,1,3cyclo-	C5H8,cyclo-
C5H10,1-pentene	C5H10,cyclo-	C5H11,pentyl	C5H11,t-pentyl	C5H12,n-pentane
C5H12,i-pentane	CH3C(CH3)2CH3	C6H2	C6H5,phenyl	C6H5O,phenoxy
C6H6	C6H5OH,phenol	C6H10,cyclo-	C6H12,1-hexene	C6H12,cyclo-
C6H13,n-hexyl	C6H14,n-hexane	C7H7,benzyl	C7H8	C7H8O,cresol-mx

C7H14,1-heptene	C7H15,n-heptyl	C7H16,n-heptane	C7H16,2-methylh	C8H8,styrene
C8H10,ethylbenz	C8H16,1-octene	C8H17,n-octyl	C8H18,n-octane	C8H18,isooctane
C9H19,n-nonyl	C10H8,naphthale	C10H21,n-decyl	C11H21	C12H9,o-bipheny
C12H10,biphenyl	HCN	HCO	HCCN	HCCO
HNC	HNCO	HNO2	HNO3	HCHO,formaldehy
HCOOH	(HCOOH)2	*N	NCO	*NH
NH2	NH3	NH2OH	NO3	N2O
NCN	N2H2	NH2NO2	N2H4	N2O3
N2O4	N2O5	N3	N3H	O3
THDCPD,endo	THDCPD,exo	N2H4(L)	C(gr)	CH3OH(L)
C2H5OH(L)	C6H14(L),n-hexa	C6H5NH2(L)	C6H6(L)	H2O(cr)
H2O(L)				

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

O/F = 2.300000

ENTHALPY	EFFECTIVE FUEL	EFFECTIVE OXIDANT	MIXTURE
(KG-MOL)(K)/KG	h(2)/R	h(1)/R	h0/R
	0.48930525E+02	0.00000000E+00	0.14827432E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.50422425E-02
*H	0.12896952E+00	0.00000000E+00	0.39081673E-01
*N	0.47845361E-01	0.21736532E-01	0.29648298E-01
*O	0.00000000E+00	0.43473063E-01	0.30299408E-01

POINT ITN	T	C	H	N	O
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Pinf/Pt = 1.727525
Pinf/Pt = 1.726019

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA
CASE = kaw00_____

	REACTANT	WT FRACTION	ENERGY	TEMP
		(SEE NOTE)	KJ/KG-MOL	K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 2.30000 %FUEL= 30.303030 R,EQ.RATIO= 0.977753 PHI,EQ.RATIO= 0.977753

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.7260	620.30
P, BAR	8.2737	4.7935	0.01334
T, K	3114.12	2965.68	1461.08
RHO, KG/CU M	7.3898-1	4.5526-1	2.7623-3
H, KJ/KG	123.28	-469.38	-4938.69
U, KJ/KG	-996.33	-1522.29	-5421.56
G, KJ/KG	-36510.2	-35356.6	-22126.3
S, KJ/(KG)(K)	11.7636	11.7636	11.7636
M, (1/n)	23.126	23.419	25.159
MW, MOL WT	23.126	23.419	25.159
(dLV/dLP)t	-1.02849	-1.02455	-1.00003

(dLV/dLT)p	1.5801	1.5270	1.0016
Cp, KJ/(KG)(K)	6.3229	6.0751	1.7637
GAMMAS	1.1280	1.1258	1.2314
SON VEL,M/SEC	1123.8	1088.7	771.1
MACH NUMBER	0.000	1.000	4.126

PERFORMANCE PARAMETERS

Ae/At	1.0000	56.394
CSTAR, M/SEC	1669.2	1669.2
CF	0.6522	1.9062
Ivac, M/SEC	2055.8	3333.6
Isp, M/SEC	1088.7	3181.8

MASS FRACTIONS

*CO	0.06726	0.05942	0.00004
*CO2	0.11623	0.12855	0.22185
*H	0.00071	0.00055	0.00000
HNO	0.00001	0.00000	0.00000
HO2	0.00007	0.00004	0.00000
*H2	0.00387	0.00333	0.00000
H2O	0.29035	0.30001	0.35193
H2O2	0.00001	0.00000	0.00000
*NO	0.01905	0.01545	0.00017
NO2	0.00002	0.00001	0.00000
*N2	0.40637	0.40805	0.41520
*O	0.00750	0.00575	0.00000
*OH	0.03913	0.03280	0.00013
*O2	0.04942	0.04602	0.01069

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH4	CH3OH	CH3OOH	*CN
CNN	COOH	*C2	C2H	C2H2,acetylene
C2H2,vinylidene	CH2CO,ketene	O(CH)2O	HO(CO)2OH	C2H3,vinyl
CH3CN	CH3CO,acetyl	C2H4	C2H4O,ethylen-o	CH3CHO,ethanal
CH3COOH	OHCH2COOH	C2H5	C2H6	CH3N2CH3
C2H5OH	CH3OCH3	CH3O2CH3	CCN	CNC
OCCN	C2N2	C2O	*C3	C3H3,1-propynl
C3H3,2-propynl	C3H4,allene	C3H4,propyne	C3H4,cyclo-	C3H5,allyl
C3H6,propylene	C3H6,cyclo-	C3H6O,propylox	C3H6O,acetone	C3H6O,propanal
C3H7,n-propyl	C3H7,i-propyl	C3H8	C3H8O,1propanol	C3H8O,2propanol
CNCOCN	C3O2	*C4	C4H2,butadiyne	C4H4,1,3-cyclo-
C4H6,butadiene	C4H6,1butyne	C4H6,2butyne	C4H6,cyclo-	C4H8,1-butene
C4H8,cis2-buten	C4H8,tr2-butene	C4H8,isobutene	C4H8,cyclo-	(CH3COOH)2
C4H9,n-butyl	C4H9,i-butyl	C4H9,s-butyl	C4H9,t-butyl	C4H10,n-butane
C4H10,isobutane	C4N2	*C5	C5H6,1,3cyclo-	C5H8,cyclo-
C5H10,1-pentene	C5H10,cyclo-	C5H11,pentyl	C5H11,t-pentyl	C5H12,n-pentane
C5H12,i-pentane	CH3C(CH3)2CH3	C6H2	C6H5,phenyl	C6H5O,phenoxy
C6H6	C6H5OH,phenol	C6H10,cyclo-	C6H12,1-hexene	C6H12,cyclo-
C6H13,n-hexyl	C6H14,n-hexane	C7H7,benzyl	C7H8	C7H8O,cresol-mx
C7H14,1-heptene	C7H15,n-heptyl	C7H16,n-heptane	C7H16,2-methylh	C8H8,styrene
C8H10,ethylbenz	C8H16,1-octene	C8H17,n-octyl	C8H18,n-octane	C8H18,isooctane
C9H19,n-nonyl	C10H8,naphthale	C10H21,n-decyl	C11H21	C12H9,o-bipheny
C12H10,biphenyl	HCN	HCO	HCCN	HCCO
HNC	HNCO	HNO2	HNO3	HCHO,formaldehy
HCOOH	(HCOOH)2	*N	NCO	*NH
NH2	NH3	NH2OH	NO3	N2O
NCN	N2H2	NH2NO2	N2H4	N2O3
N2O4	N2O5	N3	N3H	O3
THDCPD,endo	THDCPD,exo	N2H4(L)	C(gr)	CH3OH(L)
C2H5OH(L)	C6H14(L),n-hexa	C6H5NH2(L)	C6H6(L)	H2O(cr)
H2O(L)				

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

O/F = 2.400000

	EFFECTIVE FUEL h(2)/R	EFFECTIVE OXIDANT h(1)/R	MIXTURE h0/R
ENTHALPY (KG-MOL)(K)/KG	0.48930525E+02	0.00000000E+00	0.14391331E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.48939412E-02
*H	0.12896952E+00	0.00000000E+00	0.37932212E-01
*N	0.47845361E-01	0.21736532E-01	0.29415599E-01
*O	0.00000000E+00	0.43473063E-01	0.30686868E-01

POINT	ITN	T	C	H	N	O
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Pinf/Pt = 1.727537

Pinf/Pt = 1.726134

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA

CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 2.40000 %FUEL= 29.411765 R,EQ.RATIO= 0.937013 PHI,EQ.RATIO= 0.937013

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.7261	620.30
P, BAR	8.2737	4.7932	0.01334
T, K	3104.88	2956.29	1411.36
RHO, KG/CU M	7.4958-1	4.6179-1	2.8749-3
H, KJ/KG	119.66	-464.69	-4842.37
U, KJ/KG	-984.12	-1502.65	-5306.32
G, KJ/KG	-36078.3	-34930.3	-21296.5
S, KJ/(KG)(K)	11.6584	11.6584	11.6584

M, (1/n)	23.388	23.681	25.293
MW, MOL WT	23.388	23.681	25.293
(dLV/dLP)t	-1.02772	-1.02377	-1.00002
(dLV/dLT)p	1.5667	1.5125	1.0008
Cp, KJ/(KG)(K)	6.1789	5.9221	1.7228
GAMMAS	1.1280	1.1259	1.2362
SON VEL,M/SEC	1115.8	1081.1	757.3
MACH NUMBER	0.000	1.000	4.160

PERFORMANCE PARAMETERS

Ae/At	1.0000	55.121
CSTAR, M/SEC	1657.3	1657.3
CF	0.6523	1.9008
Ivac, M/SEC	2041.2	3297.5

Isp, M/SEC

1081.1 3150.2

MASS FRACTIONS

*CO	0.06084	0.05291	0.00001
*CO2	0.11978	0.13224	0.21537
*H	0.00064	0.00049	0.00000
HNO	0.00001	0.00000	0.00000
HO2	0.00007	0.00005	0.00000
*H2	0.00334	0.00283	0.00000
H2O	0.28500	0.29428	0.34161
H2O2	0.00001	0.00000	0.00000
*NO	0.02060	0.01686	0.00021
NO2	0.00002	0.00002	0.00000
*N2	0.40238	0.40414	0.41192
*O	0.00795	0.00614	0.00000
*OH	0.03977	0.03342	0.00010
*O2	0.05957	0.05661	0.03078

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH4	CH3OH	CH3OOH	*CN
CNN	COOH	*C2	C2H	C2H2,acetylene
C2H2,vinylidene	CH2CO,ketene	O(CH)2O	HO(CO)2OH	C2H3,vinyl
CH3CN	CH3CO,acetyl	C2H4	C2H4O,ethylen-o	CH3CHO,ethanal
CH3COOH	OHCH2COOH	C2H5	C2H6	CH3N2CH3
C2H5OH	CH3OCH3	CH3O2CH3	CCN	CNC
OCCN	C2N2	C2O	*C3	C3H3,1-propynl
C3H3,2-propynl	C3H4,allene	C3H4,propyne	C3H4,cyclo-	C3H5,allyl
C3H6,propylene	C3H6,cyclo-	C3H6O,propylox	C3H6O,acetone	C3H6O,propanal
C3H7,n-propyl	C3H7,i-propyl	C3H8	C3H8O,1propanol	C3H8O,2propanol
CNCOCN	C3O2	*C4	C4H2,butadiyne	C4H4,1,3-cyclo-
C4H6,butadiene	C4H6,1butyne	C4H6,2butyne	C4H6,cyclo-	C4H8,1-butene
C4H8,cis2-buten	C4H8,tr2-butene	C4H8,isobutene	C4H8,cyclo-	(CH3COOH)2
C4H9,n-butyl	C4H9,i-butyl	C4H9,s-butyl	C4H9,t-butyl	C4H10,n-butane
C4H10,isobutane	C4N2	*C5	C5H6,1,3cyclo-	C5H8,cyclo-
C5H10,1-pentene	C5H10,cyclo-	C5H11,pentyl	C5H11,t-pentyl	C5H12,n-pentane
C5H12,i-pentane	CH3C(CH3)2CH3	C6H2	C6H5,phenyl	C6H5O,phenoxy
C6H6	C6H5OH,phenol	C6H10,cyclo-	C6H12,1-hexene	C6H12,cyclo-
C6H13,n-hexyl	C6H14,n-hexane	C7H7,benzyl	C7H8	C7H8O,cresol-mx
C7H14,1-heptene	C7H15,n-heptyl	C7H16,n-heptane	C7H16,2-methylh	C8H8,styrene
C8H10,ethylbenz	C8H16,1-octene	C8H17,n-octyl	C8H18,n-octane	C8H18,isoctane
C9H19,n-nonyl	C10H8,naphthale	C10H21,n-decyl	C11H21	C12H9,o-bipheny
C12H10,biphenyl	HCN	HCO	HCCN	HCCO
HNC	HNCO	HNO2	HNO3	HCHO,formaldehy
HCOOH	(HCOOH)2	*N	NCO	*NH
NH2	NH3	NH2OH	NO3	N2O
NCN	N2H2	NH2NO2	N2H4	N2O3
N2O4	N2O5	N3	N3H	O3
THDCPD,endo	THDCPD,exo	N2H4(L)	C(gr)	CH3OH(L)
C2H5OH(L)	C6H14(L),n-hexa	C6H5NH2(L)	C6H6(L)	H2O(cr)
H2O(L)				

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS

O/F = 2.500000

ENTHALPY	EFFECTIVE FUEL h(2)/R	EFFECTIVE OXIDANT h(1)/R	MIXTURE h0/R
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(KG-MOL)(K)/KG	0.48930525E+02	0.00000000E+00	0.13980150E+02
KG-FORM.WT./KG	bi(2)	bi(1)	b0i
*C	0.16639400E-01	0.00000000E+00	0.47541144E-02
*H	0.12896952E+00	0.00000000E+00	0.36848435E-01
*N	0.47845361E-01	0.21736532E-01	0.29196197E-01
*O	0.00000000E+00	0.43473063E-01	0.31052188E-01

POINT ITN	T	C	H	N	O
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Pinf/Pt = 1.727681

Pinf/Pt = 1.726438

THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

Pin = 120.0 PSIA

CASE = kaw00_____

	REACTANT	WT FRACTION (SEE NOTE)	ENERGY KJ/KG-MOL	TEMP K
FUEL	C2H8N2(L),UDMH	0.5000000	48900.000	298.150
FUEL	N2H4	0.5000000	0.000	0.000
OXIDANT	N2O4	1.0000000	0.000	0.000

O/F= 2.50000 %FUEL= 28.571429 R,EQ.RATIO= 0.899532 PHI,EQ.RATIO= 0.899532

	CHAMBER	THROAT	EXIT
Pinf/P	1.0000	1.7264	620.30
P, BAR	8.2737	4.7923	0.01334
T, K	3093.61	2944.48	1366.75
RHO, KG/CU M	7.6024-1	4.6835-1	2.9838-3
H, KJ/KG	116.24	-460.05	-4745.33
U, KJ/KG	-972.06	-1483.29	-5192.35
G, KJ/KG	-35640.1	-34492.8	-20542.4
S, KJ/(KG)(K)	11.5581	11.5581	11.5581

M, (1/n)	23.635	23.926	25.421
MW, MOL WT	23.635	23.926	25.421
(dLV/dLP)t	-1.02668	-1.02270	-1.00001
(dLV/dLT)p	1.5482	1.4920	1.0005
Cp, KJ/(KG)(K)	6.0073	5.7338	1.6897
GAMMAS	1.1283	1.1264	1.2403
SON VEL,M/SEC	1108.1	1073.6	744.6
MACH NUMBER	0.000	1.000	4.188

PERFORMANCE PARAMETERS

Ae/At	1.00000	54.043
CSTAR, M/SEC	1645.5	1645.5
CF	0.6524	1.8950
Ivac, M/SEC	2026.7	3261.6
Isp, M/SEC	1073.6	3118.2

MASS FRACTIONS

*CO	0.05504	0.04711	0.00000
*CO2	0.12275	0.13521	0.20922
*H	0.00057	0.00043	0.00000
HNO	0.00001	0.00000	0.00000
HO2	0.00008	0.00005	0.00000
*H2	0.00290	0.00243	0.00000
H2O	0.27963	0.28853	0.33187

H2O2	0.00001	0.00000	0.00000
*NO	0.02195	0.01807	0.00021
NO2	0.00003	0.00002	0.00000
*N2	0.39868	0.40050	0.40884
*O	0.00827	0.00640	0.00000
*OH	0.04001	0.03362	0.00007
*O2	0.07006	0.06762	0.04977

* THERMODYNAMIC PROPERTIES FITTED TO 20000.K

PRODUCTS WHICH WERE CONSIDERED BUT WHOSE MASS FRACTIONS
WERE LESS THAN 5.000000E-06 FOR ALL ASSIGNED CONDITIONS

*C	*CH	CH2	CH3	CH2OH
CH3O	CH4	CH3OH	CH3OOH	*CN
CNN	COOH	*C2	C2H	C2H2, acetylene
C2H2, vinylidene	CH2CO, ketene	O(CH)2O	HO(CO)2OH	C2H3, vinyl
CH3CN	CH3CO, acetyl	C2H4	C2H4O, ethylen-o	CH3CHO, ethanal
CH3COOH	OHCH2COOH	C2H5	C2H6	CH3N2CH3
C2H5OH	CH3OCH3	CH3O2CH3	CCN	CNC
OCCN	C2N2	C2O	*C3	C3H3, 1-propynyl
C3H3, 2-propynyl	C3H4, allene	C3H4, propyne	C3H4, cyclo-	C3H5, allyl
C3H6, propylene	C3H6, cyclo-	C3H6O, propylox	C3H6O, acetone	C3H6O, propanal
C3H7, n-propyl	C3H7, i-propyl	C3H8	C3H8O, 1propanol	C3H8O, 2propanol
CNCOCN	C3O2	*C4	C4H2, butadiyne	C4H4, 1,3-cyclo-
C4H6, butadiene	C4H6, 1butyne	C4H6, 2butyne	C4H6, cyclo-	C4H8, 1-butene
C4H8, cis2-buten	C4H8, tr2-butene	C4H8, isobutene	C4H8, cyclo-	(CH3COOH)2
C4H9, n-butyl	C4H9, i-butyl	C4H9, s-butyl	C4H9, t-butyl	C4H10, n-butane
C4H10, isobutane	C4N2	*C5	C5H6, 1,3cyclo-	C5H8, cyclo-
C5H10, 1-pentene	C5H10, cyclo-	C5H11, pentyl	C5H11, t-pentyl	C5H12, n-pentane
C5H12, i-pentane	CH3C(CH3)2CH3	C6H2	C6H5, phenyl	C6H5O, phenoxy
C6H6	C6H5OH, phenol	C6H10, cyclo-	C6H12, 1-hexene	C6H12, cyclo-
C6H13, n-hexyl	C6H14, n-hexane	C7H7, benzyl	C7H8	C7H8O, cresol-mx
C7H14, 1-heptene	C7H15, n-heptyl	C7H16, n-heptane	C7H16, 2-methylh	C8H8, styrene
C8H10, ethylbenz	C8H16, 1-octene	C8H17, n-octyl	C8H18, n-octane	C8H18, isooctane
C9H19, n-nonyl	C10H8, naphthale	C10H21, n-decyl	C11H21	C12H9, o-bipheny
C12H10, biphenyl	HCN	HCO	HCCN	HCCO
HNC	HNCO	HNO2	HNO3	HCHO, formaldehy
HCOOH	(HCOOH)2	*N	NCO	*NH
NH2	NH3	NH2OH	NO3	N2O
NCN	N2H2	NH2NO2	N2H4	N2O3
N2O4	N2O5	N3	N3H	O3
THDCPD, endo	THDCPD, exo	N2H4(L)	C(gr)	CH3OH(L)
C2H5OH(L)	C6H14(L), n-hexa	C6H5NH2(L)	C6H6(L)	H2O(cr)
H2O(L)				

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS